

Essentials of Innovation and Entrepreneurship

EIE - I

Module M0 - Orientation



Agenda

- Ice Breaker – 2 Mins
- Why study Innovation and Entrepreneurship (I&E) – 3 mins
- Is I&E for Entrepreneurs only? – 4 mins
- Why is this course different? – 5 mins
- Welcome to CIE (Video and Slides) – 6 mins
- Course Overview – 6 mins
- How do we engage? – 7 mins
- Wrap Up – 7 mins

Interactive Poll

Mentimeter

Join at: www.menti.com

Code: 7880 4593



WELCOME TO ICIE

CIE History 1/2



December 2017 - CIE Starts in a small office with Prof. Sathya Prasad



January 2018 - Student Teams help design CIE's current office space

CIE History 2/2



Current Space - operational since May 2018

CIE Journey

INSPIRE

IDEATE

INNOVATE

- **1st CIE Basecamp & Bootcamp.**
- **New Courses:** CIE-Level 1 "Getting to know Entrepreneurship" & CIE-Level 2 "Building A Lean Start-up"
- **Industry Contest:** Intel India Student Contest for Game Dev & Intel Student AI/ML Contest; 3.25 lakhs in prize money
- **International Conference:** at UC Berkeley GVL 2018.
- Microsoft CodeFunDo++ 2018
- Microsoft WISE (Women in SW Engg) selects PES for inaugural Bangalore program
- **GDC-IIT Madras I-NCUBATE program:** 5 PES teams selected
- **10th Effectuation Conference, Berlin:** CIE presentation

2018-19

2020

6715
Students Graduated

1500+
Students Participating in 9 Industry Programs

- **Intel student AI/ML Contest:** 319 students, 20 Faculty, Rs 1.25 Lac in prize money
- **BV Jagadeesh (Silicon Valley VC):** 5-day Startup workshop
- **New Course:** Practical Approach to Deep Learning (with Intel AI Academy)
- UC Berkeley "Semester Abroad" program

2021

78
Industry/Startup Internships

7.4 Mil
Industry Grants/Awards

- **Cisco/NASSCOM Foundation ThingQbator Program:** 4 PES teams in top 10, Rs 20 lakhs seed funding for the 4 teams
- **LnF (Learn n Fun):** Gamified AI/ML Learning Platform (prototype), students from 3 depts
- **IEEE EDUCON (Vienna)** conference publication, co-authored with Intel India

2022

43
Cohorts Graduated

17
Capstone projects with industry partners



- **Cisco/NASSCOM Foundation ThingQbator Program Cohort 5:** FIVE PES teams in top 10, Rs 5 lakhs seed funding for each team
- CIE hosts Mr. Rostow Ravanan, Chairman, Nasscom Foundation
- AMA Talk series 'Startup Symphonies' with Startup Founders
- ICL2022 Vienna conference publication (LnF - gamified learning platform)

2023

- **CIE Aarambh program:** with Harvard Manage Mentor
- CPIP (Capstone Project with Industry Partners) kicked off with 3 companies
- **New Course:** with Intel on Firmware/ System Architecture
- Ask Me Anything (AMA) Talks series
- Invited to the IIT Bombay SemiX Advisory Board
- **CIE Summer Workshop (SWS) 2023:** a 7-week program
- **BV Jagadeesh (Silicon Valley VC):** 4-day Startup Masterclass

2024

- **EIE-I:** Pan-University course for 3rd Sem- 4500 students, 2 campuses
- **CIE Industry Mentors (IM):** 5 IMs with collective 150+ yrs of industry/start-up experience
- **CIE Innovation Project Internships (IPI):** 40 BTech interns
- **CIE Industry/Start-up Placements:** 76 Internships; 23 Full-time Offers
- **CIE Summer Workshop (SWS):** 24 Interns, XX Projects, Grants of Rs 50,000
- **Senior Industry/start-up Leader visits:** Intel, TCS, Wipro, Harman, Zoho, Microsoft, etc.
- Selected as 1 of 10 Co-Innovation Centres with IIT-D (Govt grants, Mentorship etc.)



CIE at PES University



EIE L1 - Who is Who?



Prof. Sathya Prasad
Director - CIE
Lead Instructor - EIE
sathya.prasad@pes.edu



Raghavendra Deshmukh
Industry Mentor - CIE
Co-Instructor EIE
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Mr. Madhukar Narasimha
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Mr. Tarun Rama
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Prof. Lakshmeesha
Lead Instructor - EIE (EC)
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Prof. Sathya Prasad



- **Intel (20 yrs US/India; R&D, Strategy, Prod Mgmt)**
- **New Product Innovation: Intel® Xeon™ D**
- **Founding President - SEMI India**
- **Founding Director - PESU CIE**
- **Fun Fact: Train Travel (US, Europe, Asia)**



Raghavendra Deshmukh



- SAP, Walmart, Google, Booking.com, Wipro
- Software Product & Engineering Leader, Blockchain, Cloud, Supply Chain Logistics, IoT, Product Management, Intrapreneur
- Fun Fact - Stuck in a snow fall for 5 hours in Nathu La - Sikkim



CIE Industry Mentors



**Prasanna
Chandran**

Greywiz, NTT Data

30 yrs

Cloud, Analytics, SW

- GenAI, Ai/ML
- SW Architecture
- Data Modernisation
- Computer Vision



**Nagendra
Prasad**

Wipro

30 yrs

Telco, Mobile, Auto

- Connected Car/AI
- V2V, V2X Testing
- Embedded Systems
- Computer Vision



**Raghavendra
Deshmukh**

SAP, Google

29 yrs

Software Products

- Blockchain
- AI-LLMs, RAG
- Cloud, Dist Systems
- Product Management



**Radhakrishnan
Mahalikudi**

Nvidia, AMD

35 yrs

Silicon, Semiconductors

- Processor
- Memory Arch
- SOC Arch
- Silicon Verification



**Kuldeep
Simha**

Intel

28 yrs

Silicon, Semiconductors

- VLSI design,
- CPU Design
- IP/SoC Dev
- Silicon & Systems

Fueling Innovation, Empowering Students: Industry Expertise Meets Campus Potential

EIE Teaching Assistants



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m



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Rohan R

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PESU CIE Social Media Community



[@cie.pesu](https://www.instagram.com/cie.pesu)



[PESU CIE](https://www.linkedin.com/company/pesu-cie)



[CIE On-The-Go](https://open.spotify.com/playlist/37i9dQZF1DX0XUsuxWHRQd)



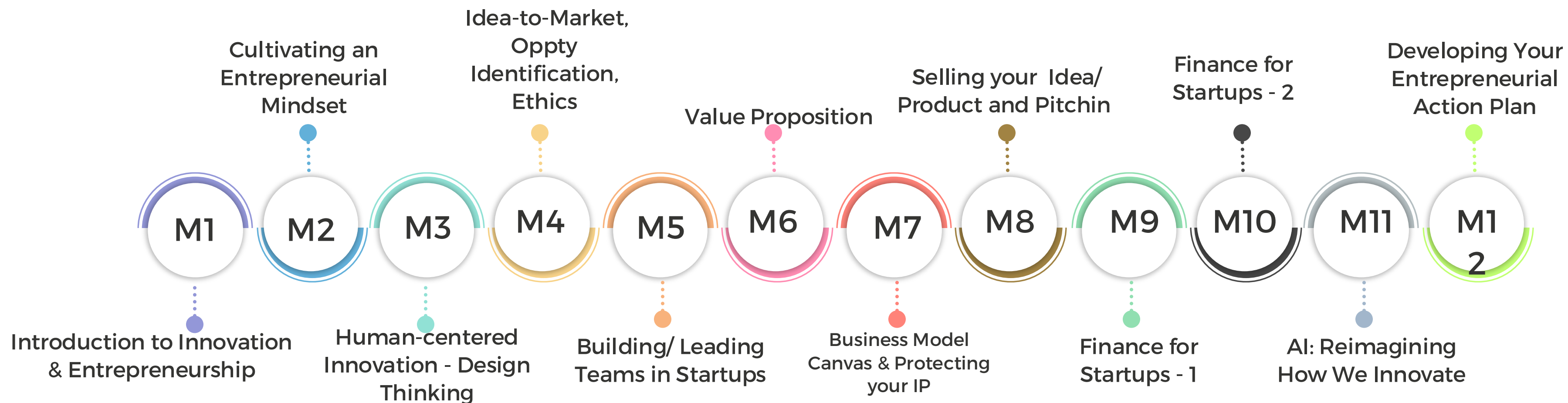
Course Objectives

- 1 Embrace I&E mindset
- 2 Kickstart your I&E journey
- 3 Understand transformative tech led changes
- 4 Business acumen
- 5 Prepare for Your Post-University Journey

Learning Outcomes

- 1 I&E Mindset Awareness
- 2 Idea-to-Market Process Familiarity
- 3 Technology Transformation Awareness
- 4 Business Fundamentals Understanding
- 5 Find like-minded students & build a network

Course Outline



Course Mechanics

- Credits: 2 Credits (Weekly 2hr)
- Engagement: Online (recommend you use all other resources: PESU, Dept & CIE)
- Will include talks by Startup Founders, Industry Experts, PESU Faculty, etc
- Grading*: S: 90 –100 A: 80 –89 B: 70.-79 C: 60 –69 D: 50 –59 E: 40 –49 F: 39 & below
- Assessment*: 2 ISAs (25% each) and 1 ESA (50%)

References

- Technology Ventures: From Idea to Enterprise, McGraw Hill
- PES University 'CIE Level 1' Course - [Getting Started With Entrepreneurship](#)
- VentureLab: [Entrepreneurial Skillset and Mindset](#)
- European Union (EU): [Entrepreneurship Competencies](#)

How to get the best of this course

- Discipline, Perseverance, Commitment
 - Key ingredients of a successful Entrepreneur
- Practice to tools/framework that you learn
 - develop skills to use the 'tools' ; not formulaic, find out what works best for you
- Lots of doing in a Team setting - You succeed with the Team
 - Entrepreneurship is a Team sport
- Digital Collaboration
 - Microsoft Teams - Chat, Discuss, Ask, Share
 - Information, Articles, Relevant News - [CIE LinkedIn](#) and CIE Instagram

Student Engagement/Facilitation

- Online
 - Q/A on MS Teams (Moderated by CIE TAs)
 - Polls
 - Session Quiz on PESU Academy - **IMPORTANT!!!**
- Offline
 - **Department Coordinator (DC)** in your Department

CIE Ignite Season 1 - Q2 2025



CIE IGNITE SEASON 01
FLAGSHIP IDEATHON BY CIE | PESU
THINK • CREATE • WIN
3000+ students, 20+ Departments,
30+ Faculty, 10+ Industry mentors, 10+ Alums

SUDEENDRA KOUSHIK
Chief Innovator & Founder
Innovation by Design,
Professor of Practice,
TEDx Speaker

BINDU SASTRY
Founder, MD, MAUVERICK
Co-founder, Director, Rasodaya
Enterprises Pvt. Ltd, BILWA
Foundation

ANAND SRI GANESH
CEO - NSRCEL, IIM
Bangalore

70K
BIGGER STAKES
BIGGER REWARDS!

1 ₹25,000
2 ₹15,000
3 ₹10,000

Pool Prize of ₹20,000

GRAND FINALE
11th April 2025 Auditorium 1B, GJB

Powered By
CO-INNOVATION CENTRE
Nurturing Excellence in Deep Tech

InspireU
Your Accountability Partner for Mentorship
PLATINUM SPONSOR

4 FOR GOOD 4GDAI
GOLD SPONSOR

IEEE COMPUTER SOCIETY
CS, PESU Bangalore Chapter
KNOWLEDGE PARTNER

Summer 2025

62
Students

7
Teaching
Assistants

**4 Industry
Mentors
(CIE-IMs)**

2
Alumni
mentor
(IISc, EY)

3
CIE Ignite
S01 teams

3
CIE Staff

Technology
Web3 Detection
Financial Degradation Generative
Scalable Remote Edge Agents
Skin VR-Based Cancer
Learning Video Computer AI Vajra
(Fintech) Vision Simulation
Interfaces Systems Deep Silicon
Brain-Computer Blockchain
RISC-V Computing
Project



Essentials of Innovation and Entrepreneurship

EIE – I

Module M1 - Introduction to Entrepreneurship

What is Entrepreneurship?

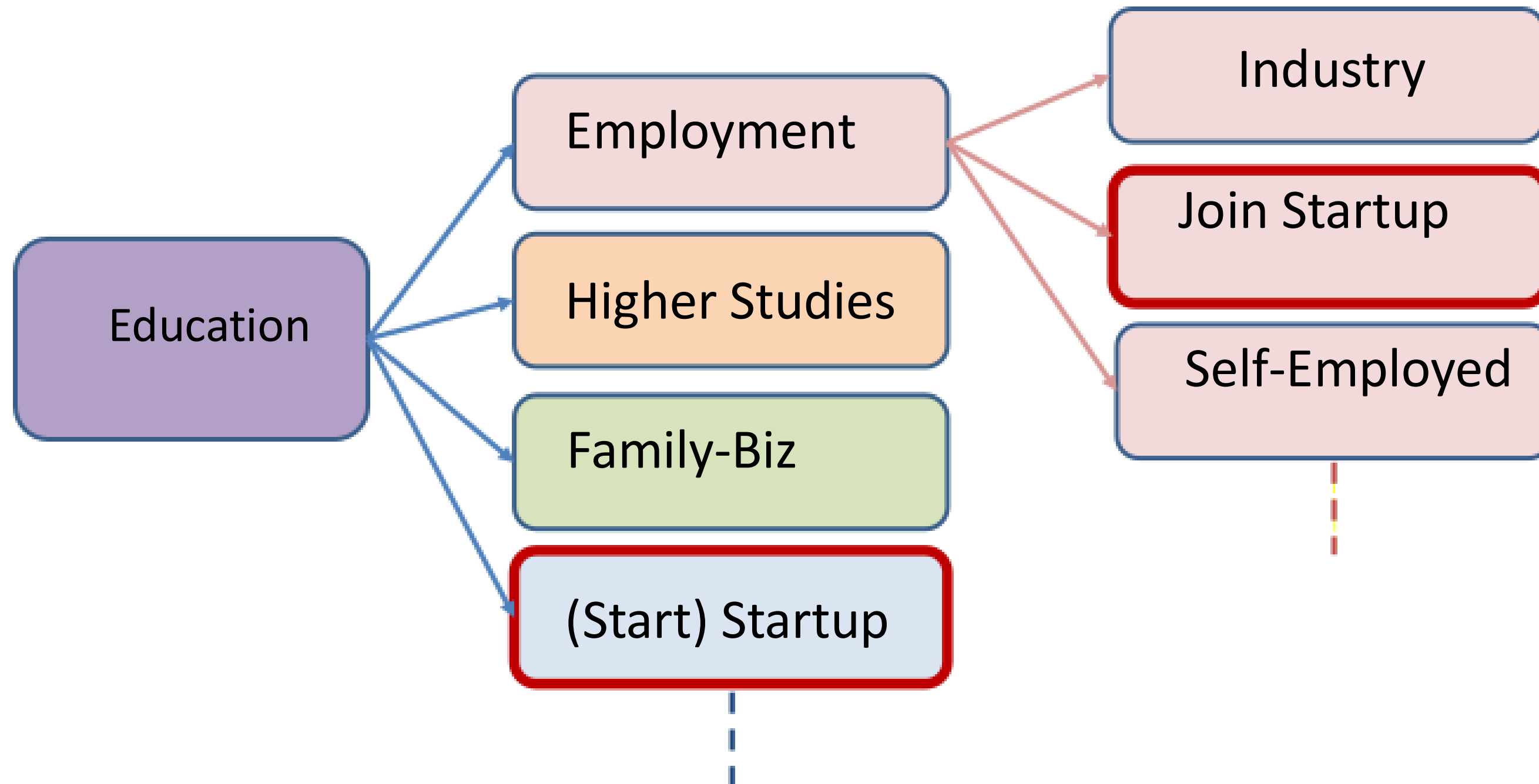
The capacity and willingness to **develop**, organize and manage a **business** venture along with any of its **risks** in order to make a **profit**.

**Entrepreneurs do much more than imaginable
with much less than seems possible***

“Entrepreneurship is when you act upon opportunities and ideas and transform them into value for others. Value created can be financial, cultural, or social”**

* Source: Stanford University ** Source: EU Entrepreneur Competence Framework

Entrepreneurial vs Entrepreneurship



Entrepreneurial vs Entrepreneurship: Example

Meet **Alex** and **Sarah** – both were Project Managers at the same company.

- Alex **stayed** in the **company** but constantly looked for **creative** ways to **improve** things.
- He took initiative, proposed bold ideas, and wasn't afraid to take risks to move the company forward.

👉 Alex is **Entrepreneurial** – he thinks like an entrepreneur, even within a company.

- Sarah, on the other hand, followed **her passion for fitness**.
- She left her job, started her own company, and began building fitness products from scratch.

👉 Sarah is an **Entrepreneur** – she took the leap to build and run her own business.

Entrepreneurial vs Entrepreneurship

What They Have in Common

- Take initiative
- Solve problems creatively
- Comfortable with risk
- Innovation-focused
- Show leadership and ownership

Aspect	Entrepreneurial	Entrepreneur
Role	Works within a company	Builds their own venture
Ownership	No business ownership	Owns and runs the business
Risk	Operates within a safety net	Takes full personal/financial risk
Resources	Uses existing company resources	Builds or raises own resources
Goal	Drive change inside	Create new value outside

Types of Entrepreneurship

Aspect	SME / MSME (Small/Micro/Medium Enterprises)	IDE (Innovation-Driven Enterprise)
Market Reach	Local	Global
Cash Flow	Lower	Higher
Risk	Lower	Higher
Profit Timeline	Gets profitable sooner	Takes longer to become profitable
Business Clustering	Less clustered	More clustered (e.g., startup hubs)

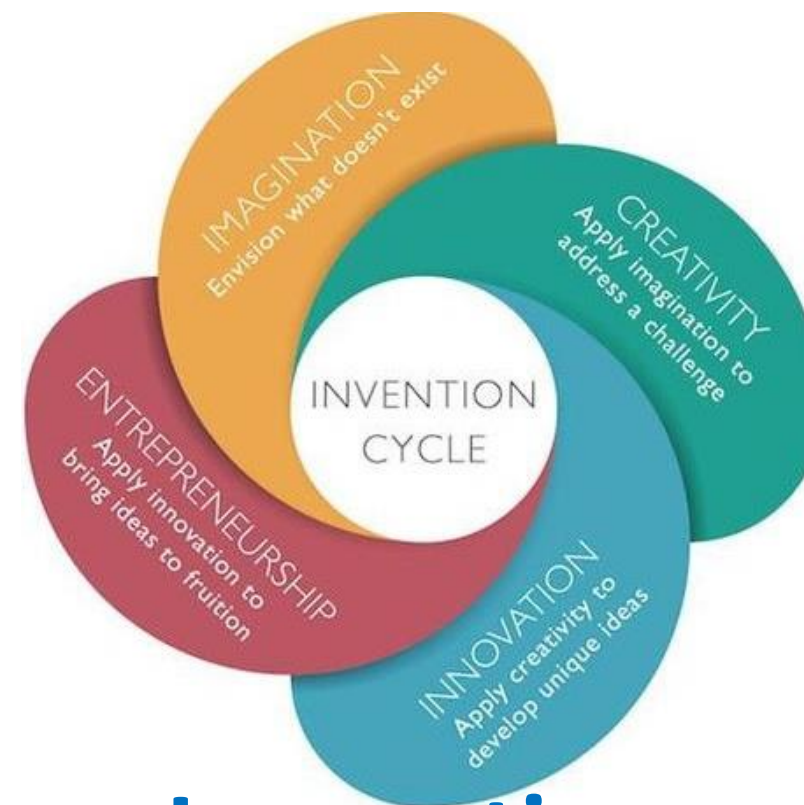
Building Blocks of Entrepreneurship

Imagination:

envisioning things that do not exist

Entrepreneurship:

applying innovation,
creating value,
thereby inspiring others'
imagination



Creativity:

applying imagination
to address a challenge

Innovation:

applying creativity to generate novel solution(s)

Imagination

Elon Musk



Elon Musk's imagination has been at the forefront of Tesla's success.
He envisioned a future where electric vehicles would be both practical and desirable

Imagination

Meta Smart Glasses:



Imagination

Examples from your life where you were
most '***Imaginative***'

Creativity



Innovation

Innovation = Invention *multiplied by* Commercialization



Innovation Elements





Entrepreneurial Myths and Traits

Myths of Entrepreneurship

- Smartest and High Achieving Individualists
- Born not made
- High risk takers
- Charismatic
- Not disciplined

Common Entrepreneurial Traits

?

Common Entrepreneurial Traits

Category	Traits
Ideas & Drive	Creativity, Vision, Spotting Opportunities, Passion
People Skills	Leadership, Influence, Building Networks, Inspiring Others
Work Style	Self-motivated, Goal-focused, Comfortable with Risk/Uncertainty, Restless
Background/Exposure	Family business, Startup experience, Friends in startups
Finance Awareness	Basic financial understanding, Budgeting, Business-savvy thinking

A Test for Entrepreneurs

The Entrepreneurial Test may assist you in your personal evaluation process. Please remember this is only a tool; this is **not** a definitive indicator whether you will be an entrepreneur or not!

Instructions for scoring for the test:
Score **3** pts for each “**Agree**” answer

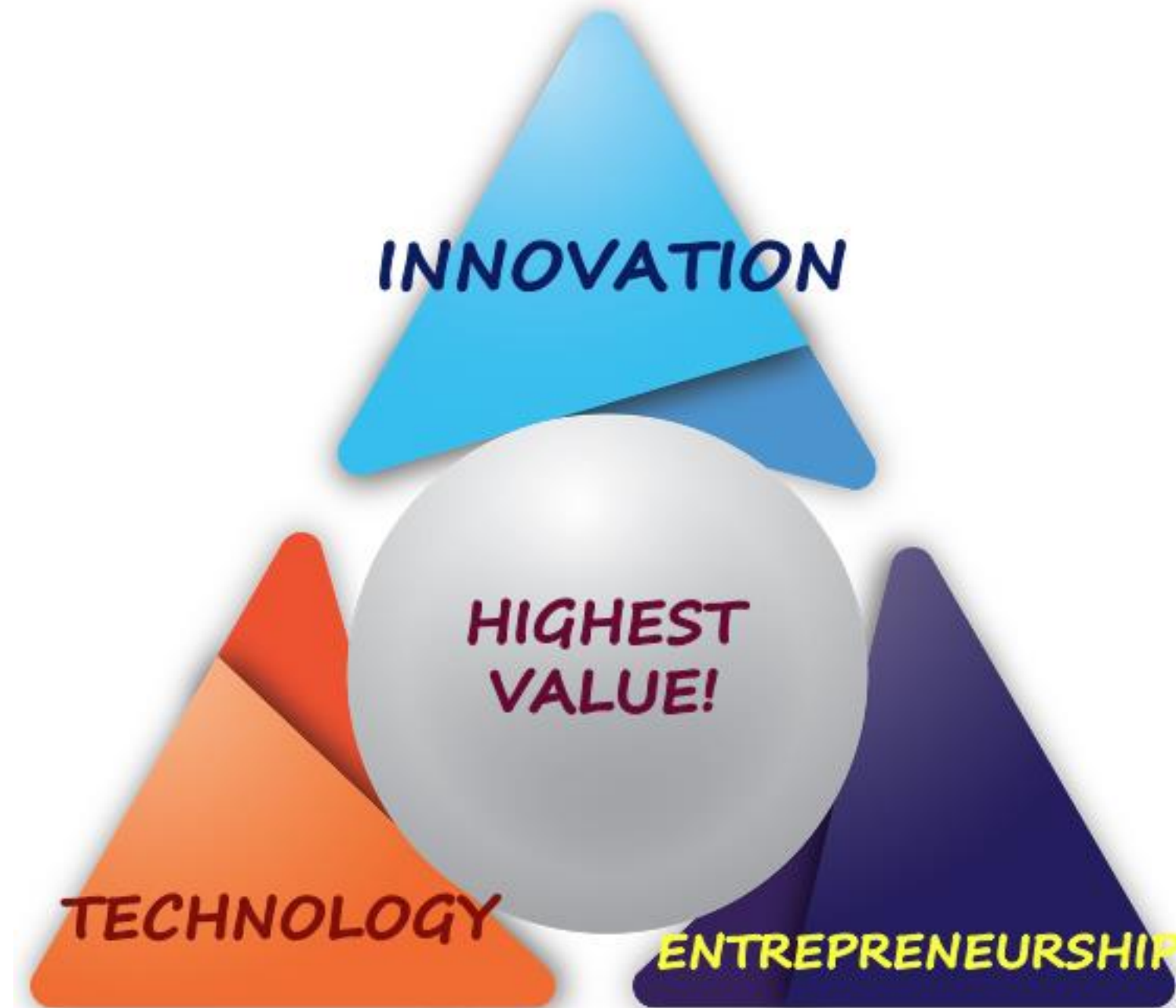
2 pts for each “**Undecided**” answer
0 pts for each “**Disagree**” answer

Startup – A Working definition

“A startup is a **temporary** organization designed to search for a **repeatable** and **scalable** business model”

VIDEO: ‘What we know about Startups’ by Kauffman Founders School [Link](#) (first 2 mins)

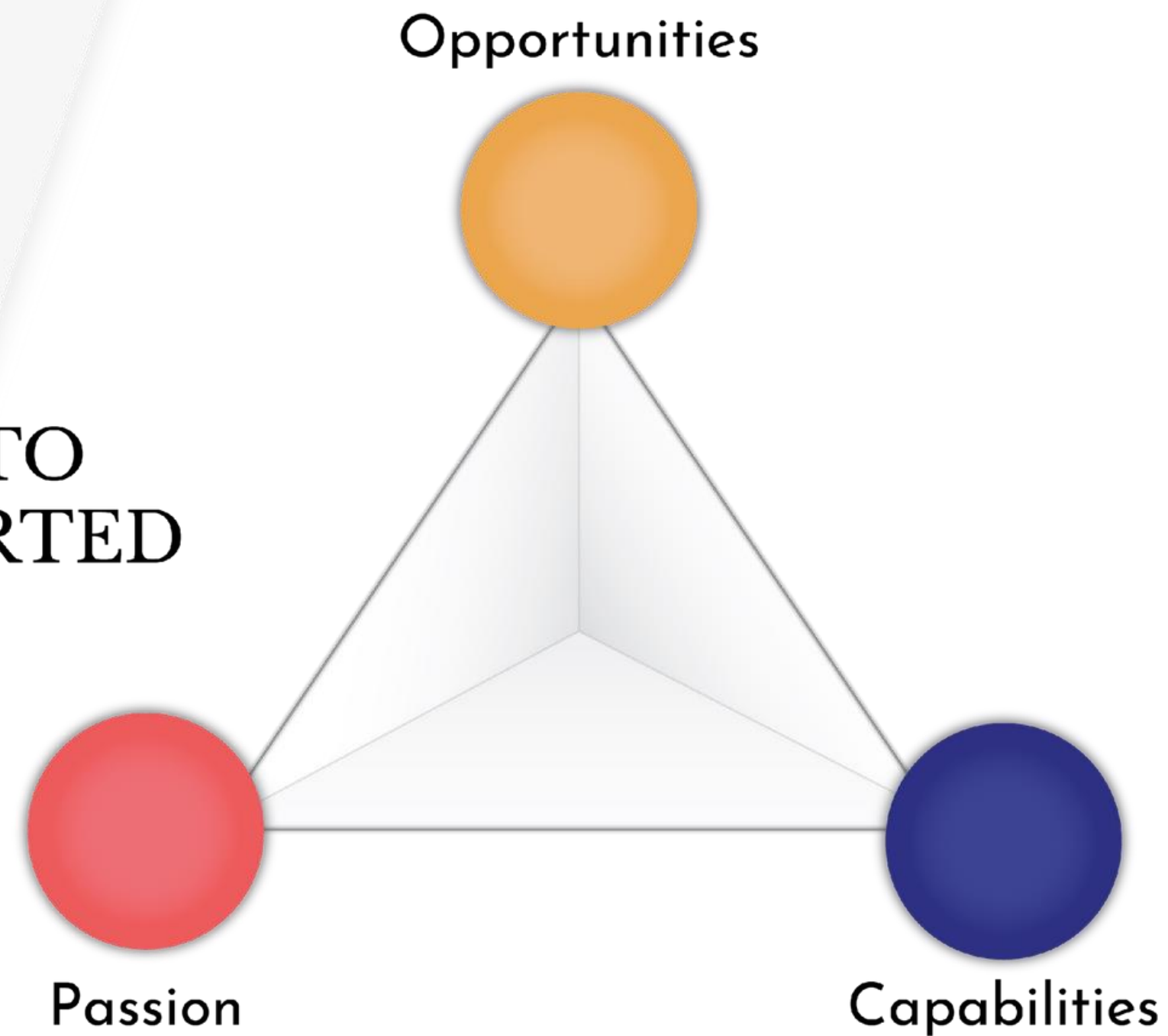
Gamechanger: Innovation & Entrepreneurship & Technology



**How will you combine Tech AND Innovation AND
Entrepreneurship ?**

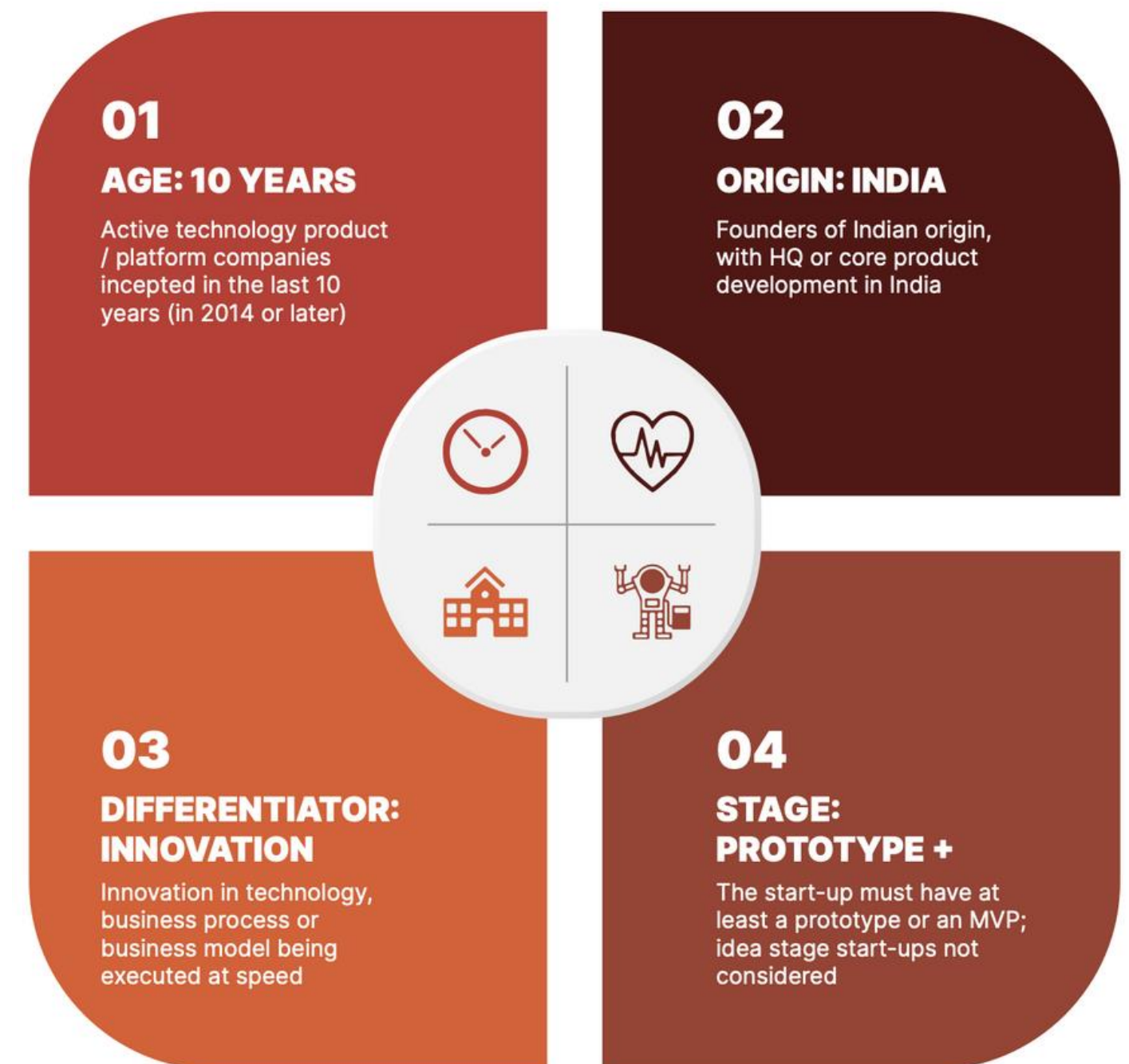
Ingredients for Entrepreneurship

3 THINGS TO
GET STARTED



Startup definition - India Perspective

“An entity working towards innovation, development, and commercialization of new products, processes, or services driven by technology or intellectual property”



[For latest Startup India [DIPP definition], pls see <https://startupindia.gov.in>]

India Startup Landscape

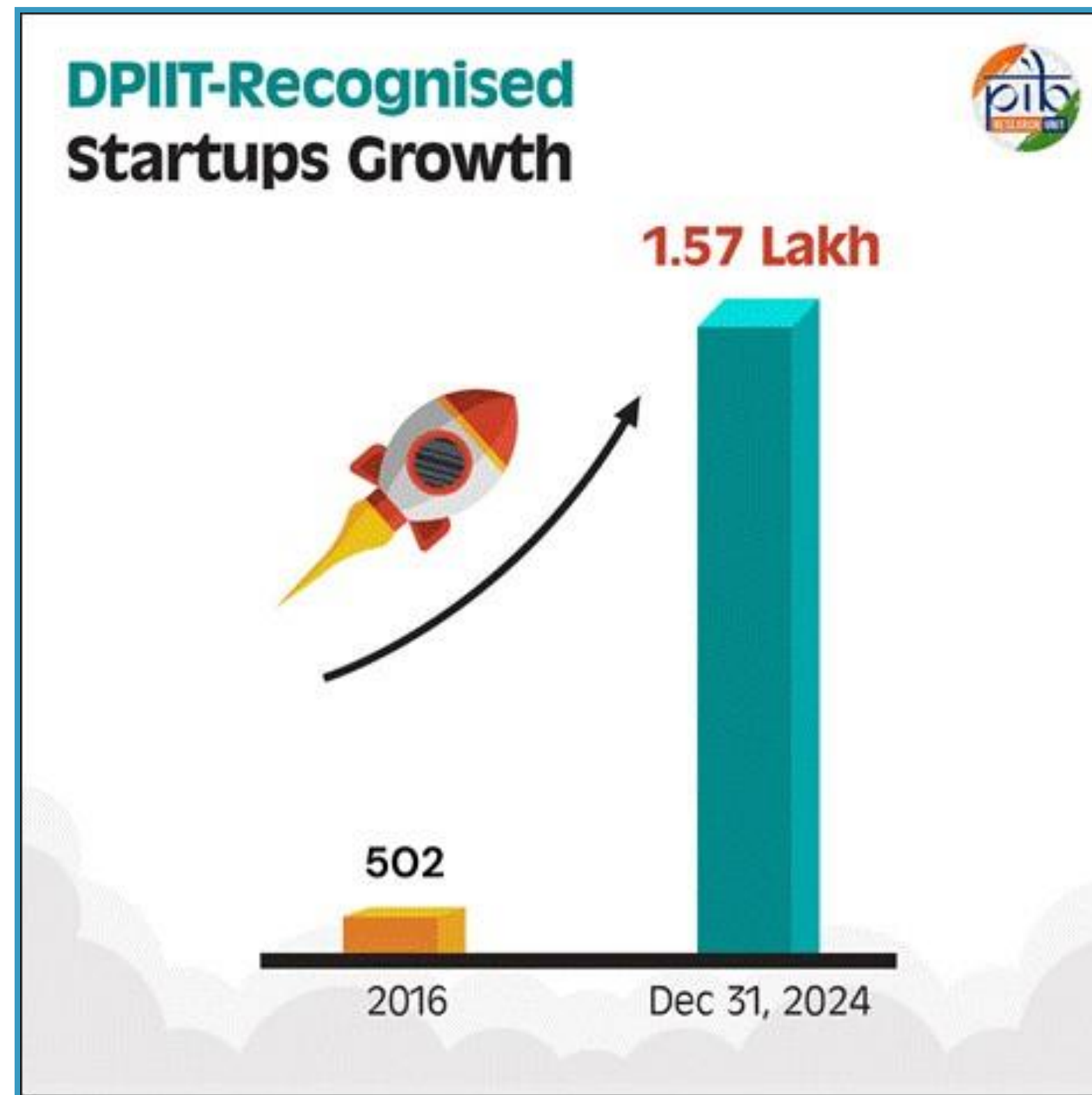
Overview of the Indian Startup Landscape

Highlights from Road to Recovery:

Indian Tech Start-up Landscape 2024 - NASSCOM-Zinnov Report

- Indian Startup Ecosystem
- Key Trends and Sectors
- Deep Tech
- And More ...

India Startup - Growth



Source: [Department for Promotion of Industry and Internal Trade \(DPIIT\)](#)

India Startup Ecosystem

Indian Tech Start-up Ecosystem: The Third-largest Globally

nasscom zinnov

	Overall	In CY 2024
Number of Active Tech Start-ups ¹	32K-35K+	2000+
Number of Unique Funded Tech Start-ups*	4000+	900+
Number of Unicorns ²	82	6
Number of Mergers & Acquisitions* ³	780+	70+
Number of Tech IPOs	27+	12
DeepTech Start-up Funding*	USD 11 Bn+	USD 1.6 Bn
Number of Funding Deals* ³	7000+	1000+

Note: (1) This report only covers Tech Start-ups founded in 2014-2024. (2) Unicorns are private entities with more than USD 1 Bn valuation. Please refer to the 'note for readers' section for detailed list of exclusions. (3) For each calendar year, funding and M&A of Start-ups founded in a 10-year timeline are taken for analysis (*) Overall numbers are for the period 2019-2024. Source: Zinnov Research & Analysis

Why India works

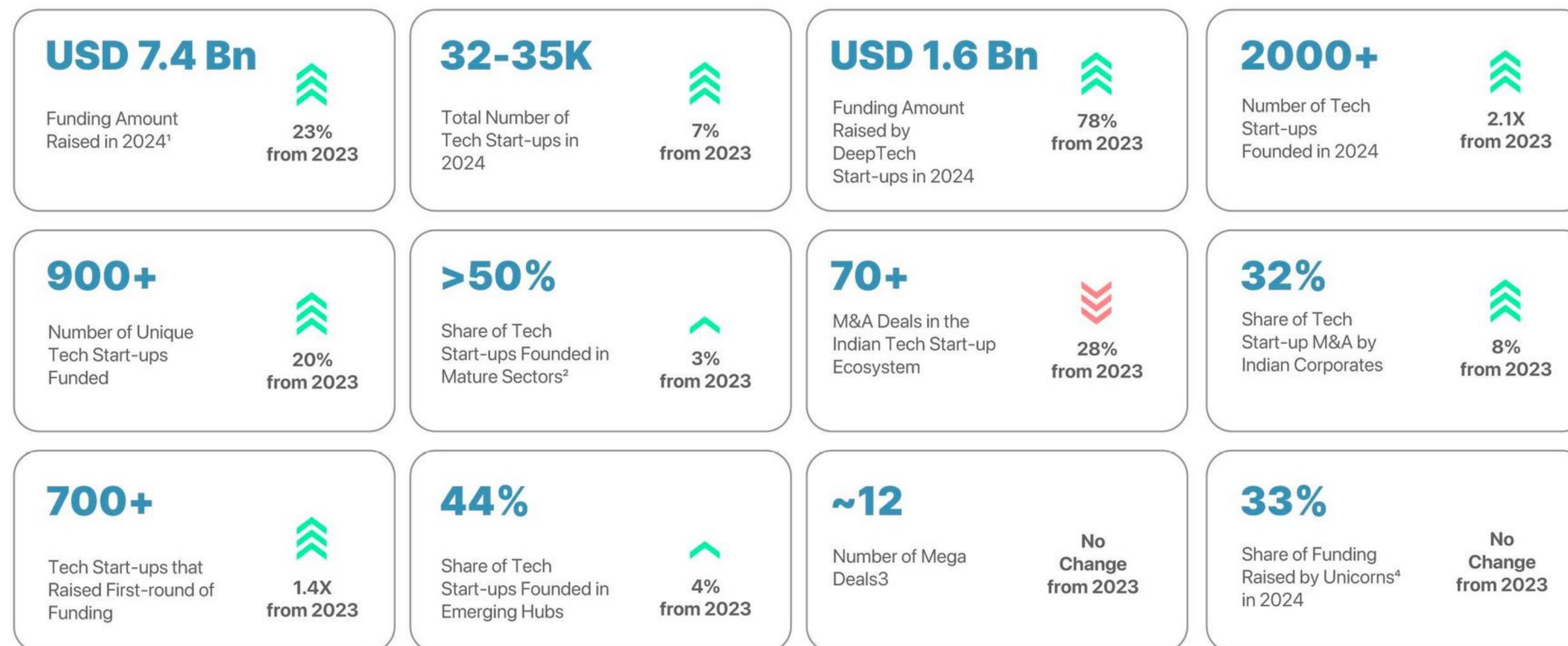


India Startup Ecosystem 2024 - By the numbers

2024 has been a year of recovery, evident in new Start-up creation, number of deals and overall investments

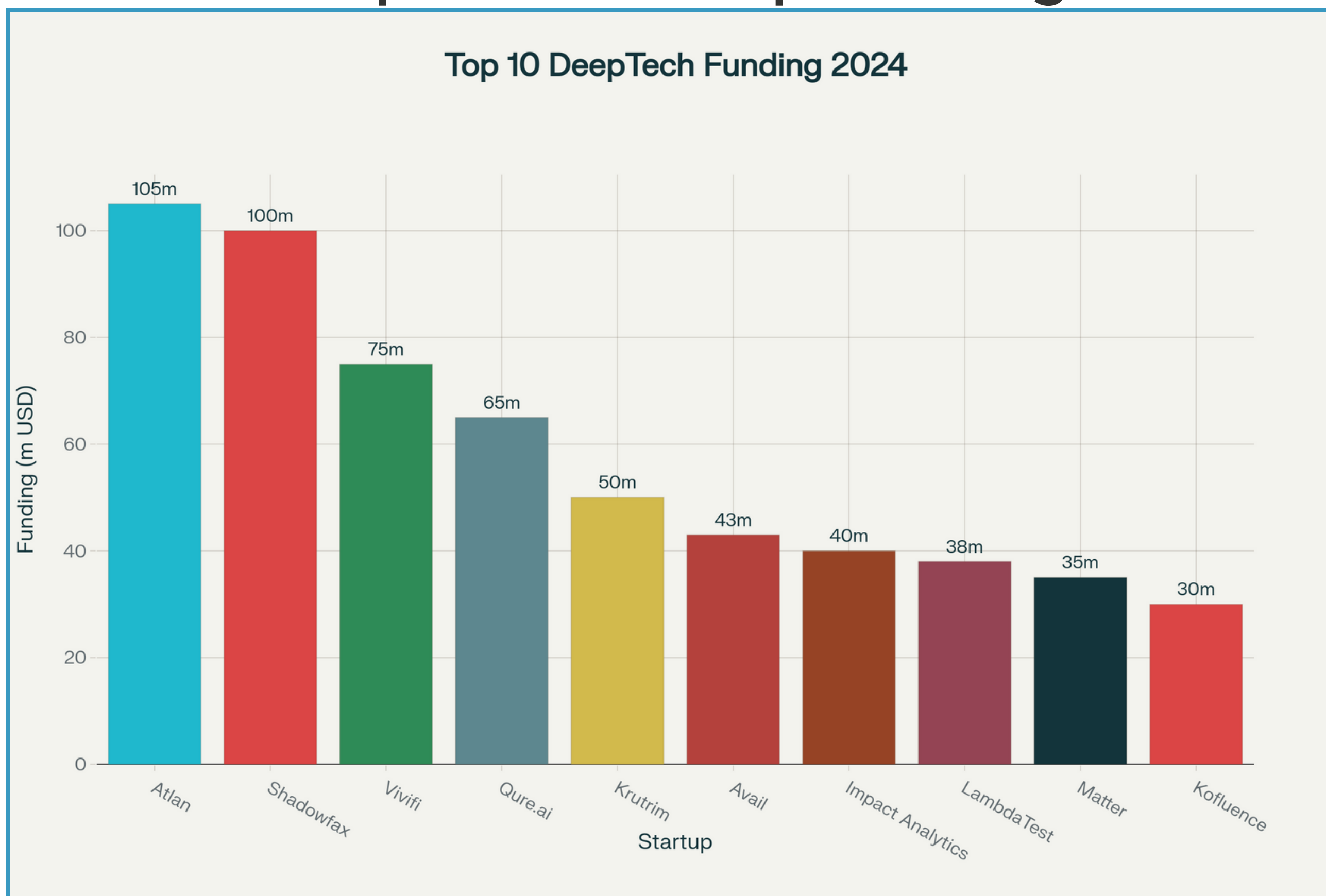
nasscom zinnov

2024 WAS A YEAR OF OUTPERFORMANCE OVER 2023 WITH VARIOUS ENCOURAGING INDICATORS OBSERVED IN THE INDIAN TECH START-UP ECOSYSTEM



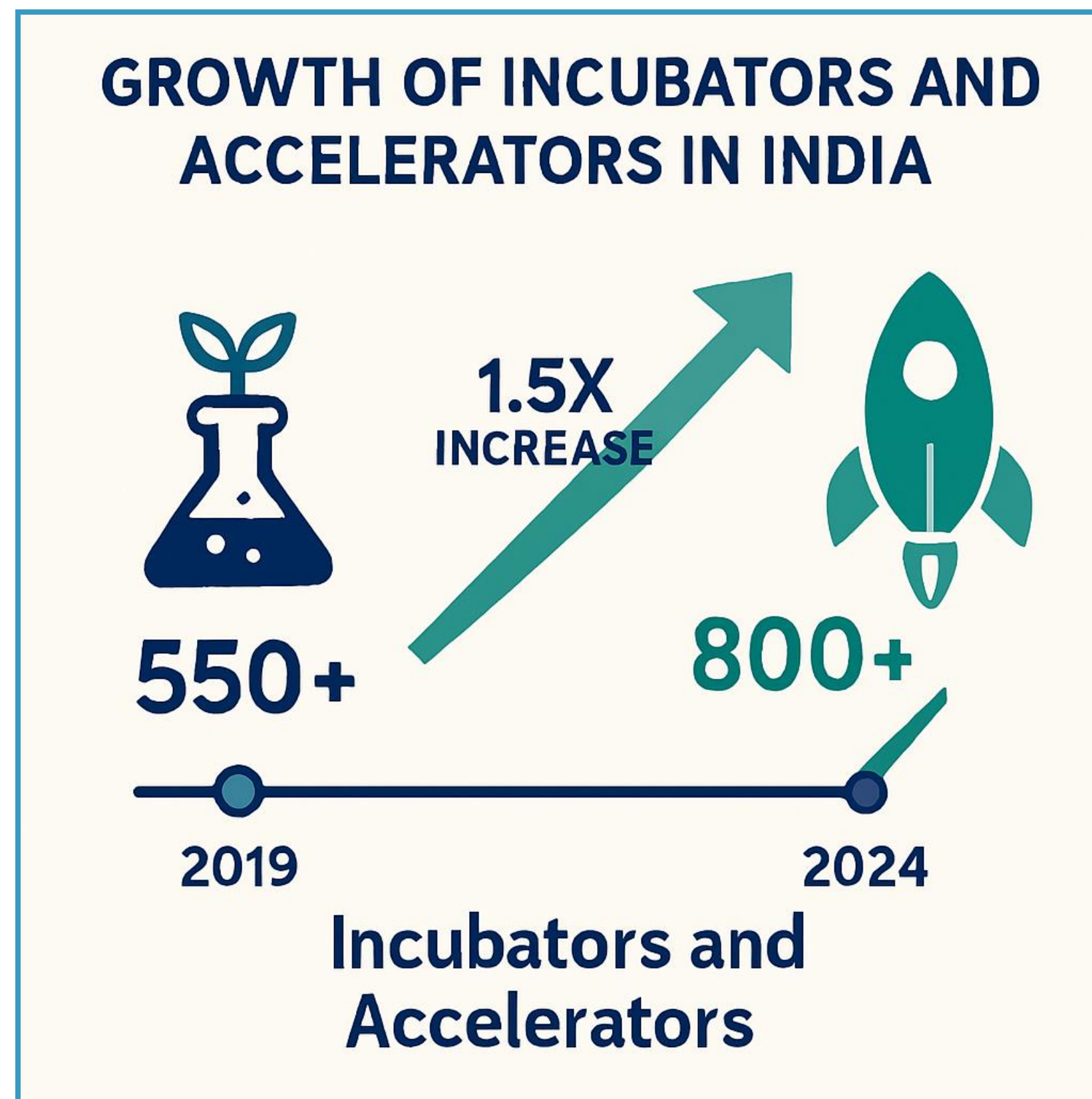
Note: (1) For each calendar year investment rounds of start-ups founded in 10-year timeline are taken for analysis, for instance for CY 2024, investment rounds of Start-ups founded between 2014-24 are considered for analysis
 (2) Mature sectors include: Edtech, EnterpriseTech, RetailTech, BFSI, HealthTech, and FoodTech. (3) Mega deals are deals with over USD 100 Mn funding (4) Refer to the "Note for readers" section. Source: Zinnov Research & Analysis

India DeepTech Startup funding in 2024



Source: NASSCOM-Zinnov-Road To Recovery: Indian Tech Startup Report 2024

Incubators and Accelerators



Non Tech Startups

Sector	Opportunities	Challenges	Examples
Healthcare	AI in diagnostics, telemedicine	Talent, regulations	Qure.ai, Niramai
Biotech	Genomics, precision health	Infra, funding gaps	String Bio, IGIB
LegalTech	Contract automation, case tracking	Low funding, niche focus	SpotDraft
Media	OTT, AI-powered content	Monetization, IP issues	Kuku FM
AdTech	Influencer tools, content targeting (AI)	Privacy, saturation	Rephrase.ai

Post Session Activity - COP Canvas

Your Photo

Your Name

Date

<u>P</u>assion	What are your interests? What 'moves' you to make the world a better place?	List 3 of your interests/passions 1. 2. 3.
<u>C</u>apabilities	Your skills/talents? Stuff (studies and beyond) that you are good at 😊	List 3 capabilities that you are proud of 1. 2. 3.
<u>O</u>pportunities	Opportunities: problems / challenges in disguise 😊	List 3 problems/challenges that you are motivated to solve 1. 2. 3.

Ikigai - A reason for being

Ikigai

(denotes 'passion that gives value and joy to life' in Japanese; [book](#))



More Information...

- CIE Office Hours by CIE Experts
- Teaching Assistants (TAs)
- CIE Ignite Season 2.0
- Reward mechanisms - Wait and Watch

Interactive Poll

Mentimeter

Join at www.menti.com
use the code: 1640 4784



Post-Session Activity

- **Work up your appetite for the upcoming weeks 😊**
- **Take a shot at the 'COP' canvas**
- **Bonus: Can you reflect on your Ikigai?**

Session Quiz

Please take the Session Quiz on PESU Academy now

Thank You!

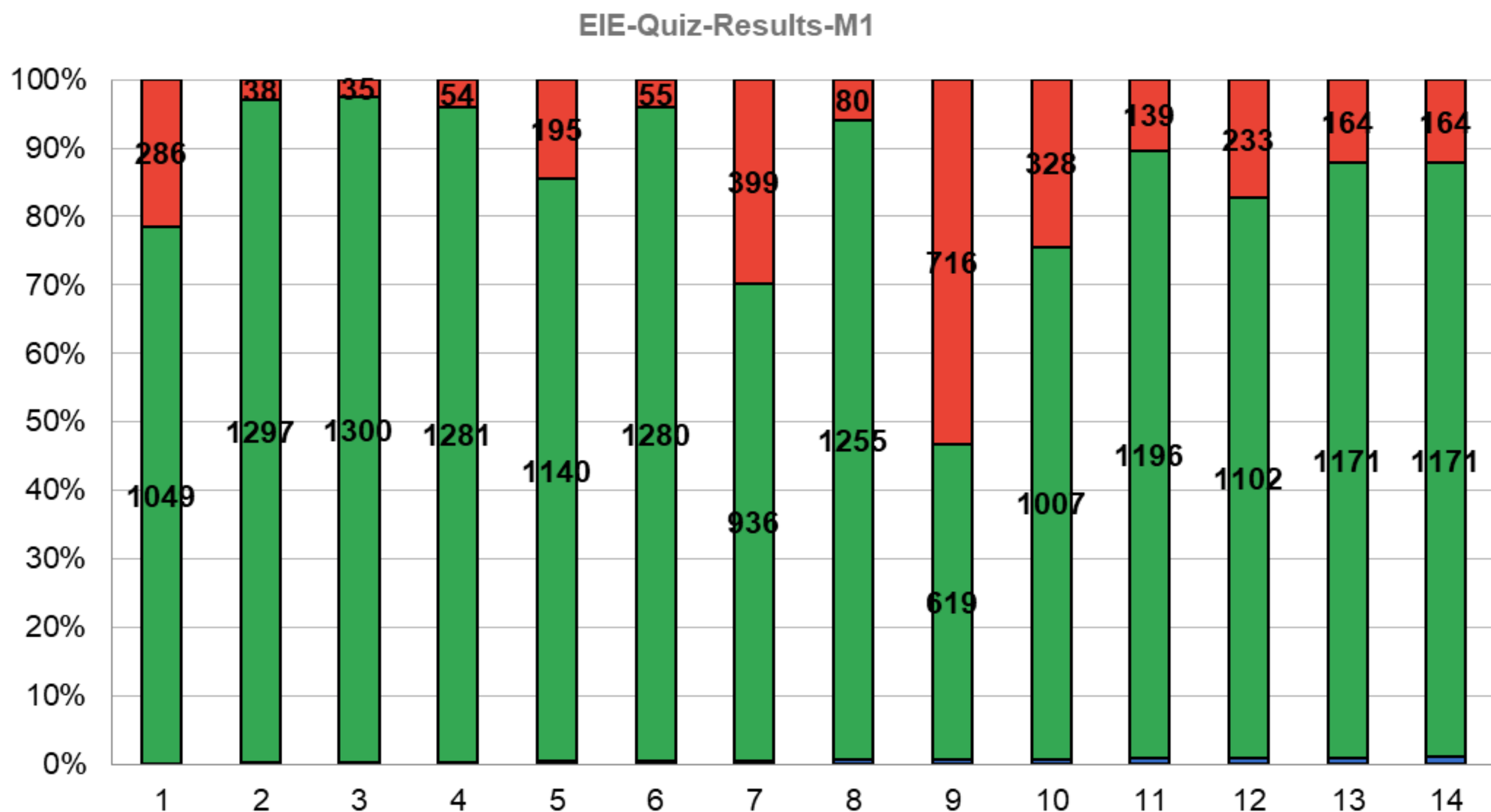
Essentials of Innovation and Entrepreneurship

EIE - I

**Module M2 - Cultivating an Entrepreneurial Mindset
+ Seeking Opportunities**

EIE M1 Session – Poll/Mentimeter Results

EIE M1 Session Quiz - Discussion



EIE M1 Session Quiz - Discussion

1.EIE is

- 1.Essentials of Innovation Engineering
- 2. Elements of Innovation and Entrepreneurship
- 3. Essentials of Innovation and Entrepreneurship: 1049 students got it right!**
- 4. East India Enterprise

9. The official/Govt definition of an Indian startup requires

- a. An idea
- b. An idea that is patentable
- c. An idea with the prospect of addressing a large market
- d. A prototype or MVP (minimum-viable-product): 619 students got it right!**

10. A game changer for a startup can be one if it combines

- a. Idea, Innovation, Technology
- b. Innovation, Technology & Entrepreneurship: 1007 students got it right!**
- c. Technology, Team, Prototype
- d. Idea, Expertise, Technology

Today's Topics

- **Prof. Sathya Prasad Quick bytes**
- **Entrepreneurial Mindset**
 - Overview
 - Growth vs. Fixed Mindset
 - Creativity, Teamwork, Adaptability, Dealing with Failure
- **Why Problems are an Entrepreneur's Best Friend!**
 - And what to do about it?
 - 'NABC' Approach

Mindset

Mindset is the **collection of beliefs, attitudes** and **ways of thinking** that *shape* how you **see the world and respond to it.**



Mindset: Growth vs. Fixed

Fixed Mindset – Belief that abilities and intelligence are static.

Growth Mindset – Belief that abilities can be developed through effort, learning, and persistence.

Fixed Mindset

- Abilities are fixed
- Avoids challenges
- Gives up easily
- Ignores feedback
- Threatened by others' success

Growth Mindset

- Abilities can grow
- Seeks challenges
- Persists through setbacks
- Learns from feedback
- Inspired by others' success

Growth vs. Fixed Mindset



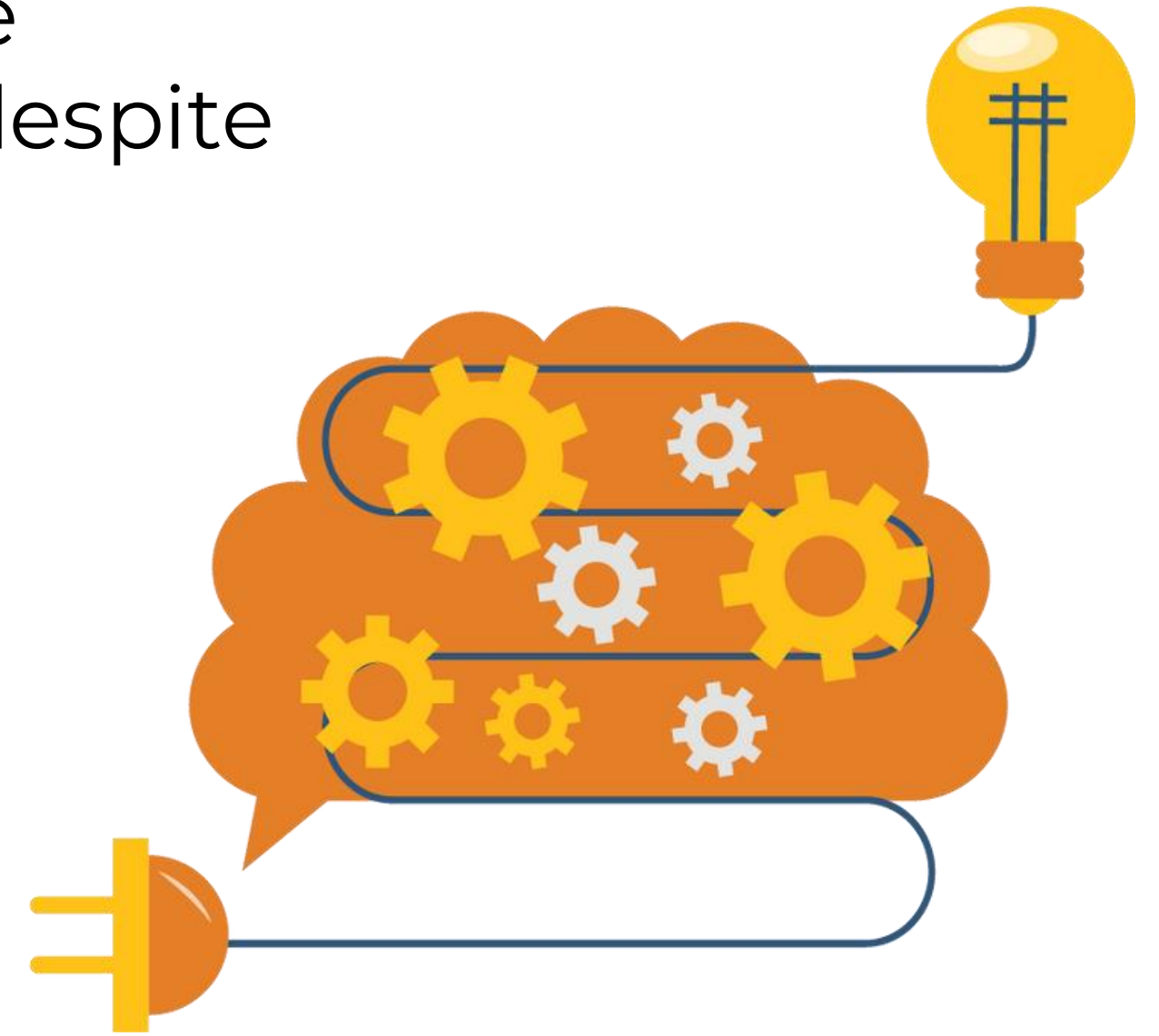
Also watch Video: [Developing a Growth Mindset with Carol Dweck](https://www.youtube.com/watch?v=hiiEeMN7vbQ) (10m)

Video Source: <https://www.youtube.com/watch?v=hiiEeMN7vbQ>

Entrepreneurial Mindset

A way of thinking that enables you to recognize opportunities, take initiative, and create value despite uncertainty.

- Entrepreneurial Mindset (EM) is learnable
- You can condition your attitude
- How: Get Started x Practice x Time



**Your entrepreneurial
journey begins here!**



Entrepreneurial Mindset:

The Heart of Entrepreneurship



The Heart of Entrepreneurship

Just like the heart pumps life into the body, Entrepreneurial Mindset pumps drive, creativity, and resilience into every entrepreneurial journey.

The Common Thread

Whether it's a tech startup, a social enterprise/NGOs, a family business, or corporate innovation—Entrepreneurial Mindset is the *same engine* powering action, adaptability, and opportunity recognition.

Entrepreneurial Mindset – Why does it matter?



Video Credits: <https://www.youtube.com/watch?v=WZHvRpuemgk>

Entrepreneurial Mindset Cards

A Catalog with Prompt Questions

Self-Activity

- What Actions will **you** take?
- Test yourself again after some time to see the change



Entrepreneurial Mindsets



Always ask "why?" and "what if..." to seek new and creative ways of learning and doing.



Recognize your potential to learn and accomplish anything you work hard to achieve



Dare to step out of your comfort zone to accomplish great things



Learn to work through failure to create solutions for the problems you are passionate about



Actively observe, identify, and act on circumstances where you can innovate or solve a problem



Proactively develop creative solutions to the problems around you



Learn to work through difficult moments and view failure as an opportunity to grow and learn something new



Be confident and hopeful in your ability to innovate and your potential for making an impact



Develop quick and clever ways to overcome challenges (and don't worry, you can always iterate!)



Think about other people's needs and feelings keeping these in mind when solving a problem



Make smart changes when challenges arise without losing heart or giving up

Entrepreneurial Skills

1. CREATIVITY

Being creative means being bold and imaginative and not limiting your thinking to what's already done

2. TEAMWORK

It's rare for someone to accomplish things alone. Seek out teams with diverse perspectives, skills, and talents

3. PUBLIC SPEAKING

Recognize your potential to learn and accomplish anything you work hard to achieve

4. IDEA GENERATION

The key to innovation is to let your mind spread its wings and fly. The more ideas the better, wild and silly ideas welcome!

5. OPPORTUNITY ANALYSIS

Entrepreneurs look for different patterns and trends to see if an idea will be successful and if it's the right time to get started

6. MARKET RESEARCH

It's important to gather information about your potential users so you can turn your ideas into something they will want

7. DESIGN THINKING

To develop effective solutions, entrepreneurs need to put the user first and seek out feedback in order to design with users in mind

8. PROTOTYPING

Entrepreneurs create prototypes--simple models to explain their idea, get feedback, and learn how their product can be improved

9. BUSINESS MODEL

Business models help answer questions like: What will it cost to make your product? What should you change?

10. PITCHING

Rallying people around your idea is critical! Effective "pitches" to potential users and investors are clear, short, and persuasive

Deep Dive: Let's Pick a Few...

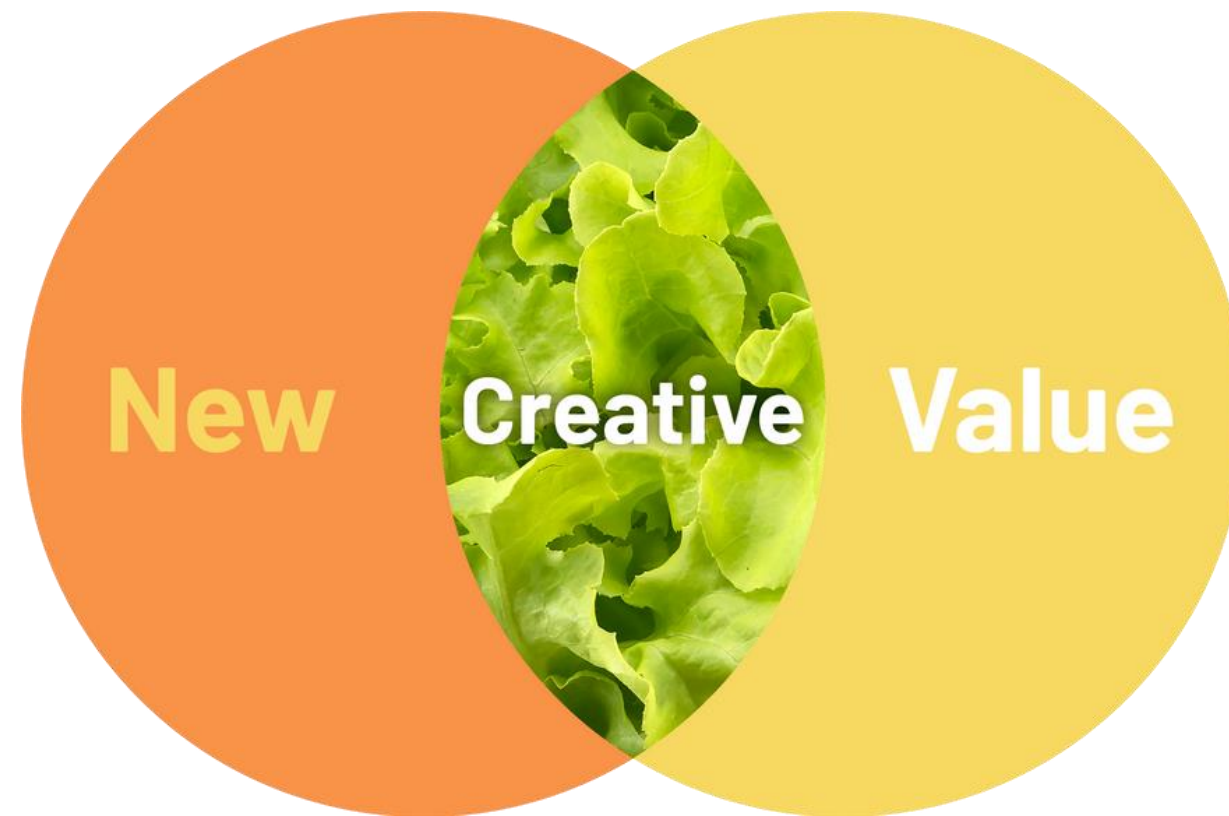
Creativity

Adaptability

Dealing with Failure

Teamwork

Creativity



Creativity lies at the **intersection** of projects that bring new ideas and value addition together.

Example:

- A new feature makes things easier
- A service that meets a need better
- Improving what users already have

Video: [The Art and Science of Creativity - Tom Kelley at TEDxTokyo](#)
(13m)

Teamwork

Coordinated effort of individuals working toward a common goal.

Key Role in Success: In entrepreneurship, innovation, and projects, *no single person* can do everything—teams multiply skills, perspectives, and energy.

Outcome: Strong teamwork turns diverse individuals into a *high-performing unit*.

“A **team** is like a jazz band —different instruments, different notes but when they listen and adapt to each other, the result is magic.”



Team Stages

Tuckman's Stages of Team Development - Music Analogy (with Adjourning)

Forming:
Introduction stage
Polite, exploring roles
□ Musicians meet, tune instruments

Storming:
Conflict stage
Differences cause friction
□ Disagree over tempo/arrangement

Norming:
Alignment stage
Trust and processes form
□ Agree on arrangement, find groove

Performing:
High-performance stage
Smooth, goal-focused
□ Full flow, effortless performance

Adjourning:
Closure stage
Wrapping up, parting ways
□ Final bow, packing instruments

Video: Teamwork related



Simon Sinek:
What Makes a Team Great?
(2:30m)



Steve Jobs:
Managing people (2:30m)



NBA:
Teamwork San Antonio Spurs
(2:30m)

Adaptability

The skill of **recognizing** when your **assumptions** no longer **match reality** — and having the **courage** and **creativity** to shift **direction** without losing **momentum**.

In Entrepreneurial Mindset, adaptability means **embracing uncertainty** as a constant, **quickly learning** from feedback, and **pivoting** when necessary to **create value** in changing environments.



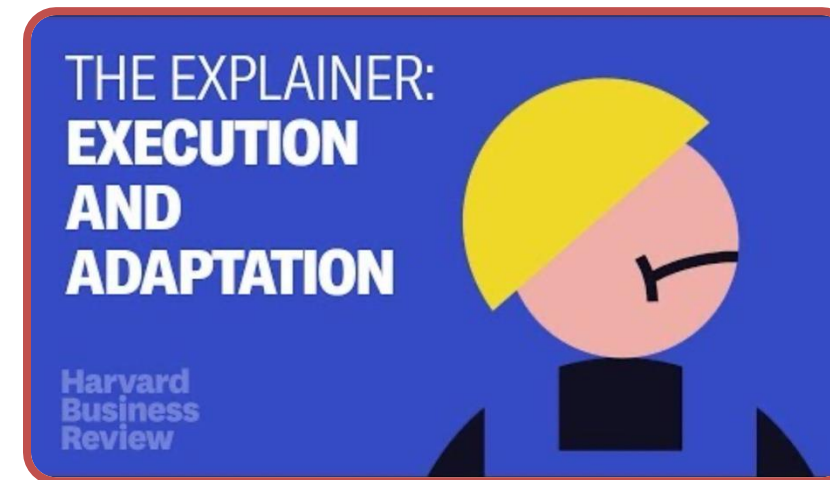
Videos



3 ways to measure your adaptability and how to improve it
| Natalie Fratto



How to Adapt to Changing Times
Simon Sinek



HBR Explainer:
Balancing Execution and Adaptation

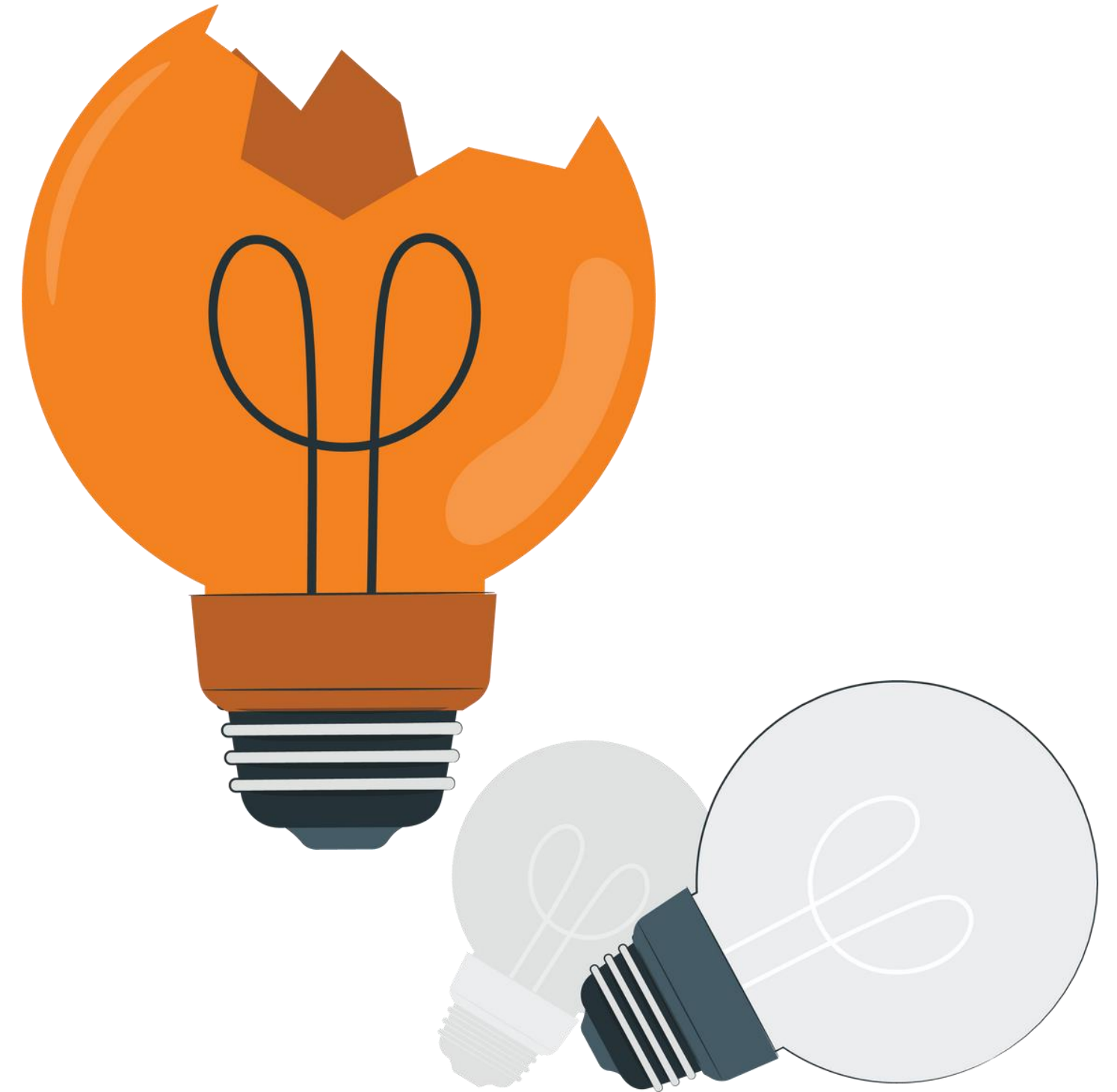


Andrew Grove, "Strategic Inflection Points" -
1996 MIT Industry Leaders Program Lecture

Dealing with Failure

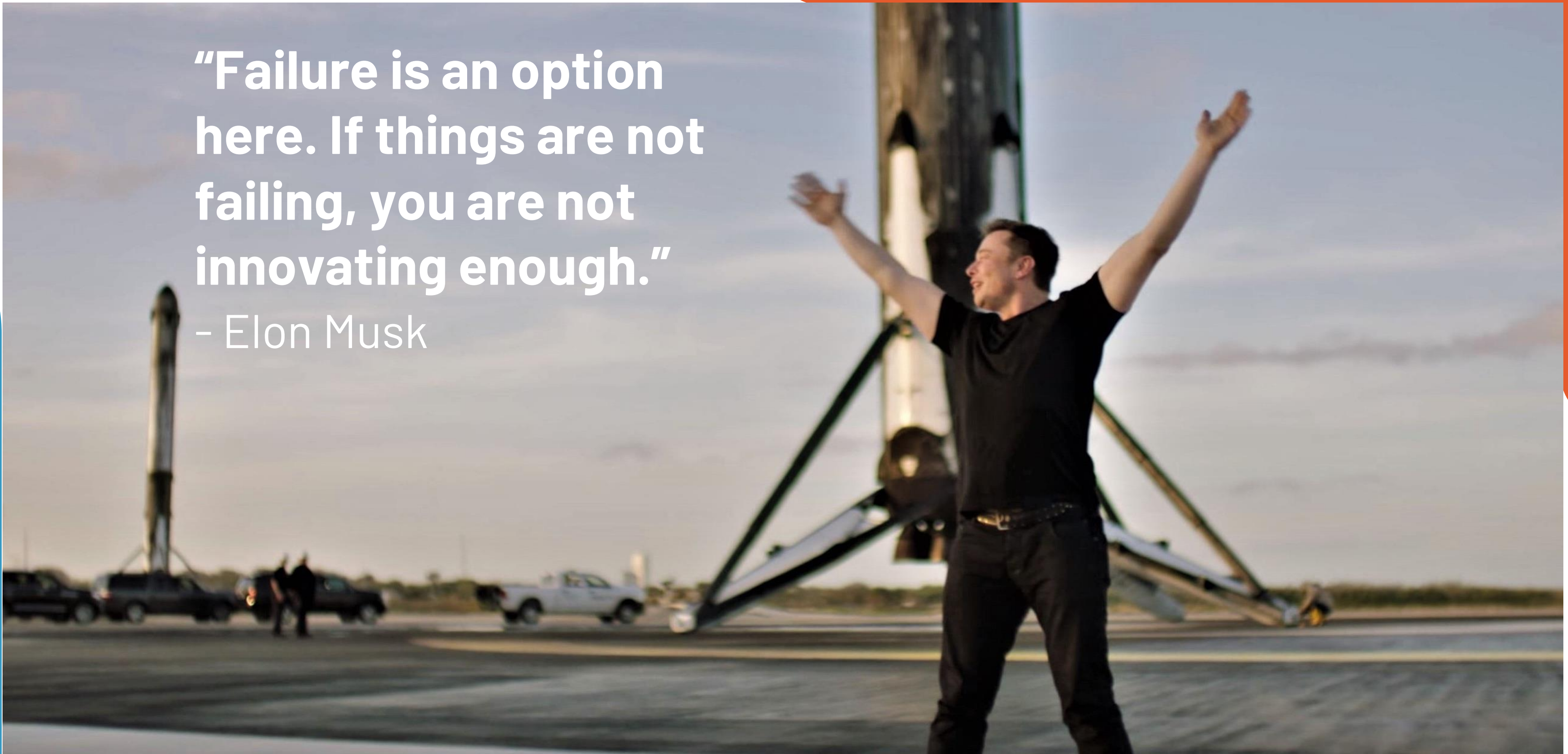
Failure, the unexpected teacher that guides entrepreneurs through the *maze of innovation*.

It's the catalyst for growth,
the spark that ignites
resilience and adaptability.



“Failure is an option here. If things are not failing, you are not innovating enough.”

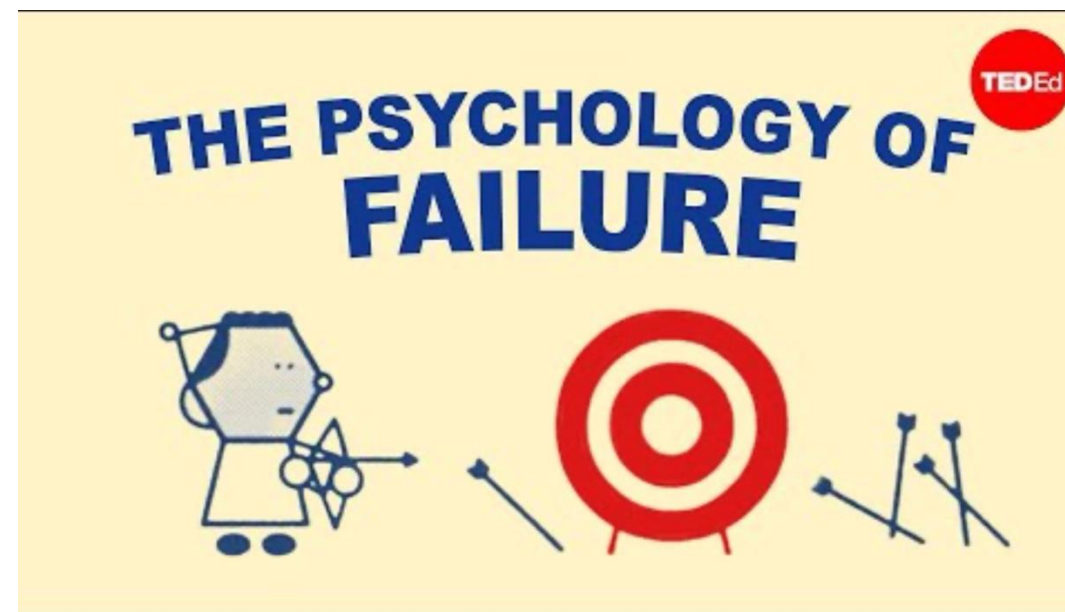
– Elon Musk



Video: Dealing with Failure



Elon Musk on Failure
Tesla (0:25m)



How to Overcome your Mistakes
Ted-Ed (4:51m)



"The Fringe Benefits of Failure" J.K.
Rowling
Harvard Commencement speech (21m)

https://www.youtube.com/shorts/5_eW76S9Mkl
<https://www.youtube.com/watch?v=eBz7iUJu9UM>
<https://youtu.be/wHGqp8lz36c>

What do all these people have in common?



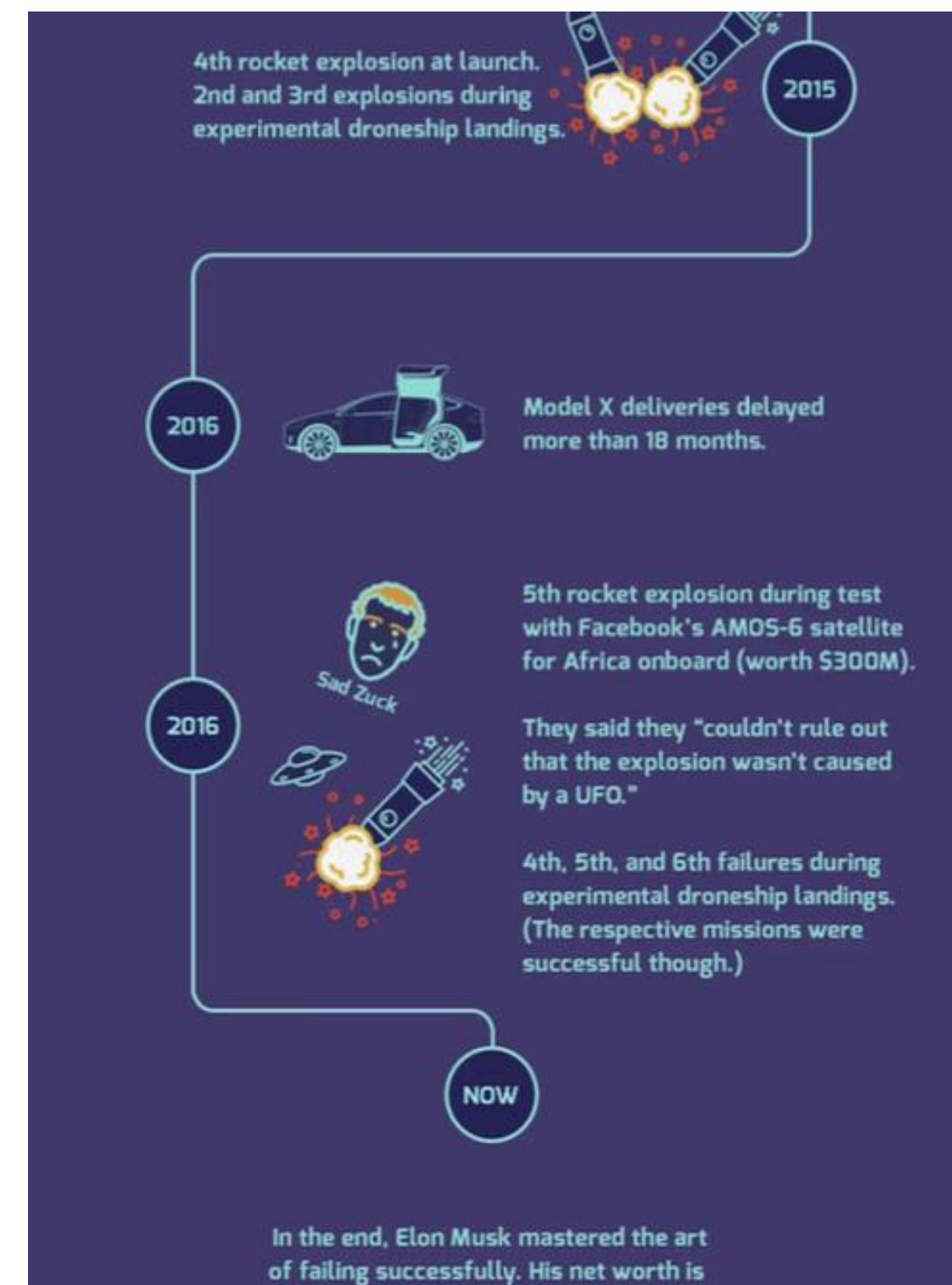
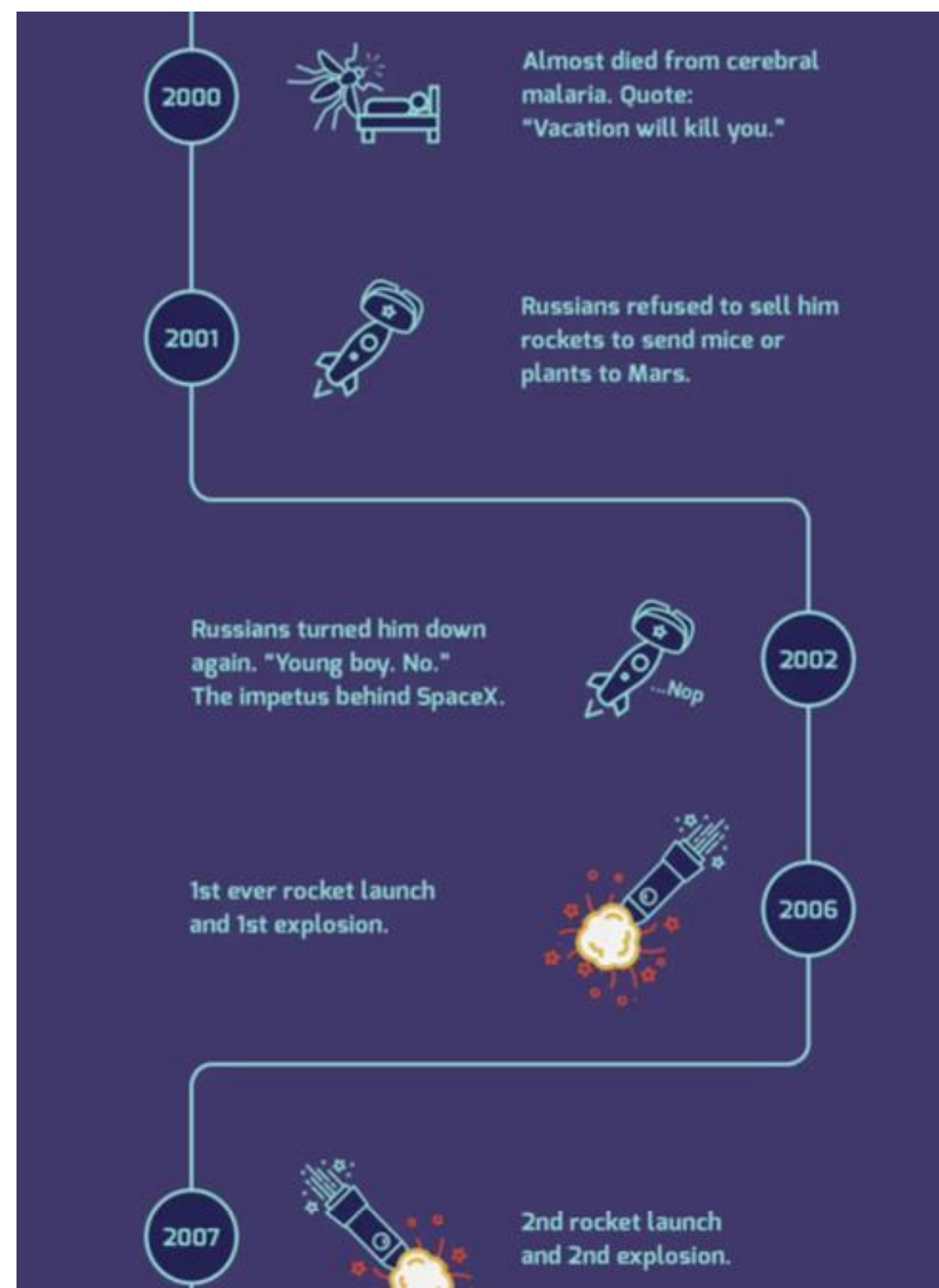
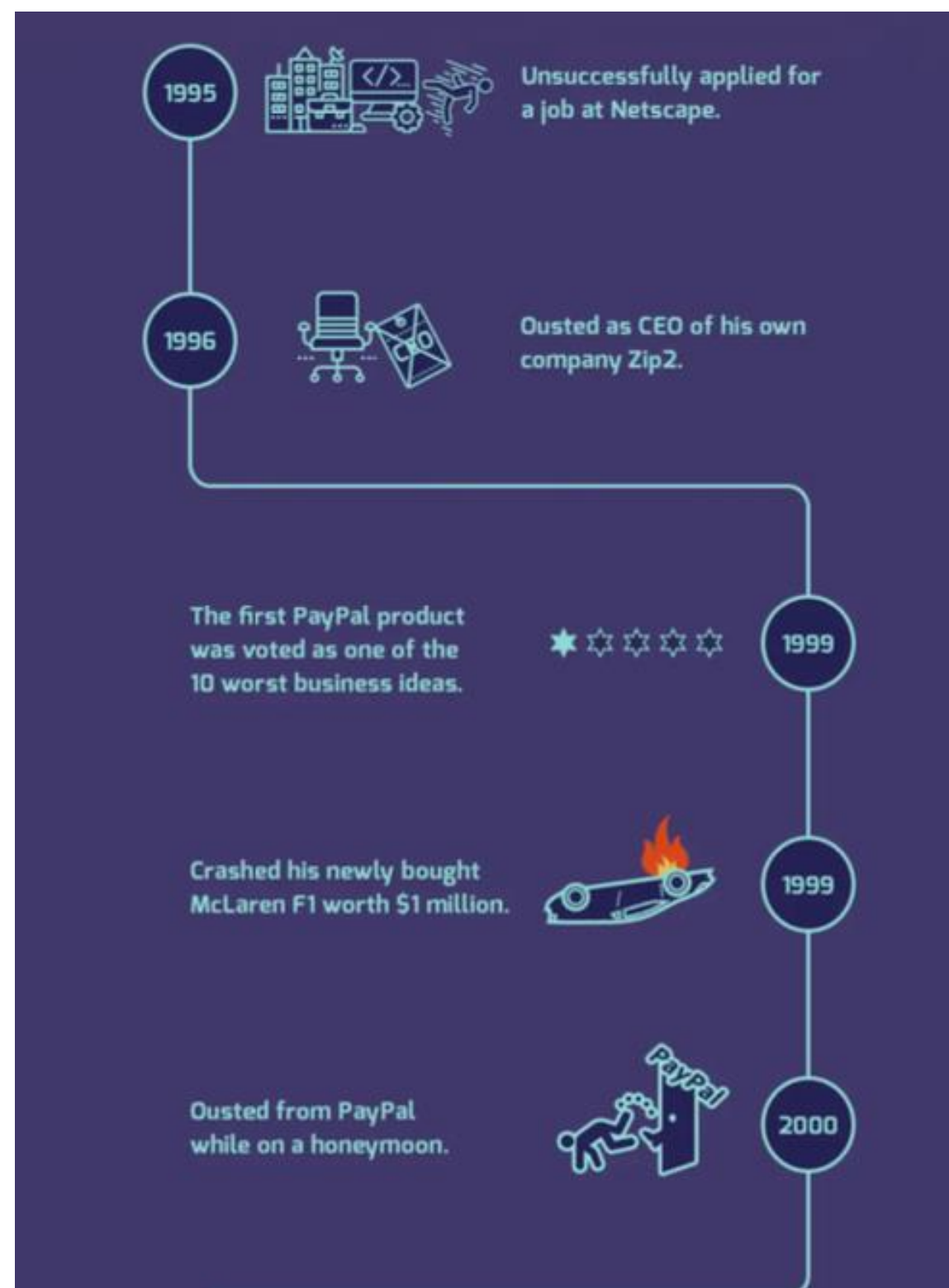
What do all these people have in common?



**They overcame
crushing failures
before becoming
extremely
successful!**



Elon Musk's 'Failure Resume'



Dealing with Failure

“The difference between average people and **achieving** people is their perception and response to **failure**.”

- John Maxwell

Post-Session Activity – Failure Resume

- Can you come up with your own ‘Failure Resume’?

Capture 3 (total) of your biggest failures across:

- School/Education
- Sports
- Competition(s)
- Personal Goals

- **For each of your failures, capture your response to below questions:**

- My Biggest Failure Taught Me...
- And Changed My Behavior By....
- I Will Reflect On My Next Failure By ...

**"Ask the Founders"
(live interview)
Ananya Mungara – SmartChakra
ananya@smartchakra.co**

Founder Talk

Ananya Mungara - SmartChakra

Profile

Name & Designation: Ananya Mungara – CEO & Founder, SmartChakra Technologies Pvt. Ltd.

Professional Overview:

CEO & Founder of **SmartChakra Technologies**, a pioneering deep-tech company delivering advanced predictive maintenance, condition monitoring, and smart asset management solutions for the automotive and manufacturing sectors. With expertise spanning **Technology, Operations, Sales, and Marketing**, we have led the end-to-end journey of two flagship innovations:

- **SONICSCAPE** – an industrial monitoring solution integrating multi-sensor data (vibration, temperature, audio) with AI-driven analytics to enable predictive maintenance, real-time condition monitoring, and quality assurance.
- **SmartChakra Tyre Management System** – an intelligent tyre health monitoring and inflation-deflation-puncture management platform designed for fleet operators, heavy vehicles, and industrial equipment to improve safety, efficiency, and cost savings.

Key Roles & Expertise:

- Leading **technology innovation**, from ideation to large-scale deployment, ensuring solutions are practical, scalable, and future-ready.
- Overseeing **operations**, ensuring smooth delivery and integration of technology across diverse client environments.
- Driving **sales and strategic partnerships** with leading automotive and manufacturing OEMs, securing high-impact deployments.
- Spearheading **marketing and brand positioning** to establish SmartChakra as a trusted industry technology partner in both industrial monitoring and tyre management solutions.

Achievements & Recognitions:

- Holder of **multiple patents** in the fields of sensor-based industrial monitoring, predictive analytics, and smart tyre management.
- **Guest speaker** at multiple nationally and internationally recognized technology and industry forums.
- Won the best paper award at IEEE, Malaysia, won the Youngest Achiever Award by Rotary, Won the National Wonder Woman Award by Lions International and many more..
- Opportunity to discuss with distinguished Indian leaders, including:
 - **Nirmala Sitharaman, Dr. S. Jaishankar, Dr. Rajeev Chandrasekhar** and many more..

Industry Contributions:

- Serving as a **technology partner** for leading automotive and manufacturing OEMs, enabling data-driven decision-making.
- Driving **digital transformation** in legacy manufacturing environments by integrating smart analytics platforms.
- Innovating in the mobility sector with the **SmartChakra Tyre Management System**, helping industries and fleets improve road safety, reduce maintenance costs, and optimize tyre performance.



Profile

Name & Designation: Ananya Mungara – CEO & Founder, SmartChakra Technologies Pvt. Ltd.

Professional Overview:

CEO & Founder of SmartChakra Technologies, a pioneering deep-tech company delivering advanced predictive maintenance, condition monitoring, and smart asset management solutions for the automotive and manufacturing sectors. With expertise spanning Technology, Operations, Sales, and Marketing, we have led the end-to-end journey of two flagship innovations:

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Guest speaker at multiple nationally and internationally recognized technology and industry forums

Picture

Best way to Generate Ideas?

*Don't Look for Ideas,
Look for Problems!!*

*Most of the time, we
overlook many problems...*

Memo at Western Union (1878)

“This ‘telephone’ has too many shortcomings to be seriously considered as a means of communication. The device is inherently of no value to us.”



Business Week (1968)

“With over 50 foreign cars on sale here, the Japanese auto industry isn’t likely to carve out a big slice of the US market.”



Leo DeForest Radio Pioneer & Inventor of the Vacuum Tube (1926)

“While theoretically and technically, television may be feasible, commercially and financially it is an impossibility, a development of which we need waste little time dreaming.”



Charles Duell, Commissioner, U. S. Office of Patents (1899)

“Everything that can be invented has been invented.”



UNITED STATES PATENT
AND
TRADEMARK OFFICE

So, What Can you Do?

Post-Session Activity: Before Dismissing a Problem

- Next time you come across a problem and want to dismiss it, can you try:
 - Is there some aspect of the problem/idea you like?
 - Are there other problems you think worth solving that are similar to this problem?
 - What are some assumptions that made you dismiss the problem?
Can you test those assumptions?
- 3 powerful words
 - **“How Might We”**
 - **“How might we** design a reusable rocket to reduce cost of the space program?
 - **“How might we** enable people without credit card to buy using e-commerce?
 - **How might we** build a foundational Indic LLM model at a fraction of the cost?

Can you guess the companies that may have asked the above questions?

Capturing a Problem/Idea/Opportunity Statement: NABC* Approach

*An important client or market need addressed
by a unique approach with compelling benefits
when compared against the competition or alternatives.*

* Created by Stanford Research Institute (SRI) – Read the 2-pager [here](#)

Post-Session Activity: NABC Framework

For a problem capture the proposed problem & solution using the NABC framework (internet offers many more/other templates as well)

Project:
Team:
Version & Date:



NABC

Quick guide: The NABC (Need, Approach, Benefit, Competition) is the minimal form of a business idea structuring method. This method is used to document or generate ideas and to present business ideas and innovation projects to feedback providers.

More tips & tricks for this template on book page: 177



Lewrick / Link / Leifer
The Design Thinking Toolbox
978-1-119-62919-1

1 Name and description of the idea	
Need ♡ <i>Describe the need</i> 2	Approach ⇒ ♡ <i>Describe the approach</i> 3
Benefit ♡ + <i>Formulate the benefit</i> 4	Competition ♡ 🏆 <i>Describe alternatives and/or competitors and unique selling propositions</i> 5

 THE DESIGN THINKING TOOLBOX

BASIC TEMPLATE

Get a PDF
Premium Design
Thinking Template:
WWW.DT-TOOLBOOK.COM/SHOP

Session Feedback

Mentimeter

Join at www.menti.com
use the code: 6746 1723



Session Quiz

Please take the Session Quiz on PESU Academy now

Thank You!

Startup Nation Olympiad 2025

Why PESU EIE Students Should Participate?

What's In It For You?

- **Global Exposure:** Pitch your ideas to an audience of over 1 crore students, 1000+ universities, and 500+ incubation centres.
- **Mentorship:** Get guidance from experienced founders and domain-specific mentors through Startup Nation's curated network.
- **Collaboration:** Build teams, find co-founders, and collaborate with peers from diverse backgrounds.
- **Marketplace Access:** List your startup products/services for free and reach industry partners, potential customers, and investors.
- **Job Opportunities:** Explore the Startup Jobs portal to join early-stage teams or recruit talent for your venture.
- **Funding & Support:** Compete for access to the Startup Nation Fund and connect with investors who fuel bold ideas.



<https://www.startupnationindia.com>

Startup Nation Olympiad 2025

Why PESU EIE Students Should Participate?

Student-Led Startups

- Fellowship Up To ₹10 Lakhs To Top 100 Student Teams
- Conditional Term Sheet Up To ₹2 Crore
- Assistance With Company Registration, IP Filing And Trademark
- Network With Industry And Research Organizations For Product Development

Why Participate?

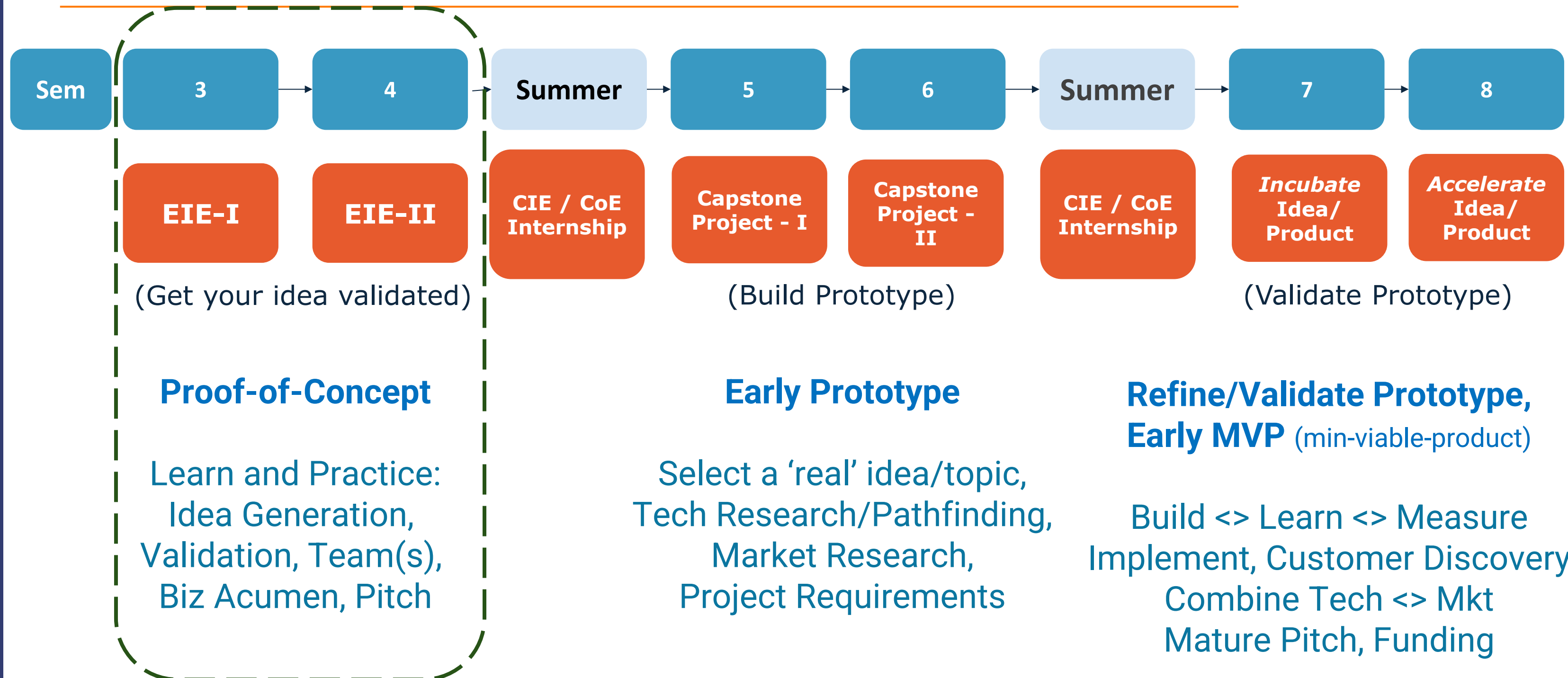
- **Real-World Experience:** Apply your classroom learning to real startup challenges.
- **Skill Development:** Enhance leadership, teamwork, and problem-solving skills.
- **Validation:** Get early feedback, validate your ideas, and accelerate your startup journey.
- **Network:** Expand your professional connections and learn from top founders.

How To Get Started?

- Register Now: www.startupnationindia.com
- Prepare: Use available guides/FAQs to get ready for the competition.
- Engage: Tap into resources, mentorship, and training modules to maximize your impact.

<https://www.startupnationindia.com>

Your Innovation Journey Starts NOW!



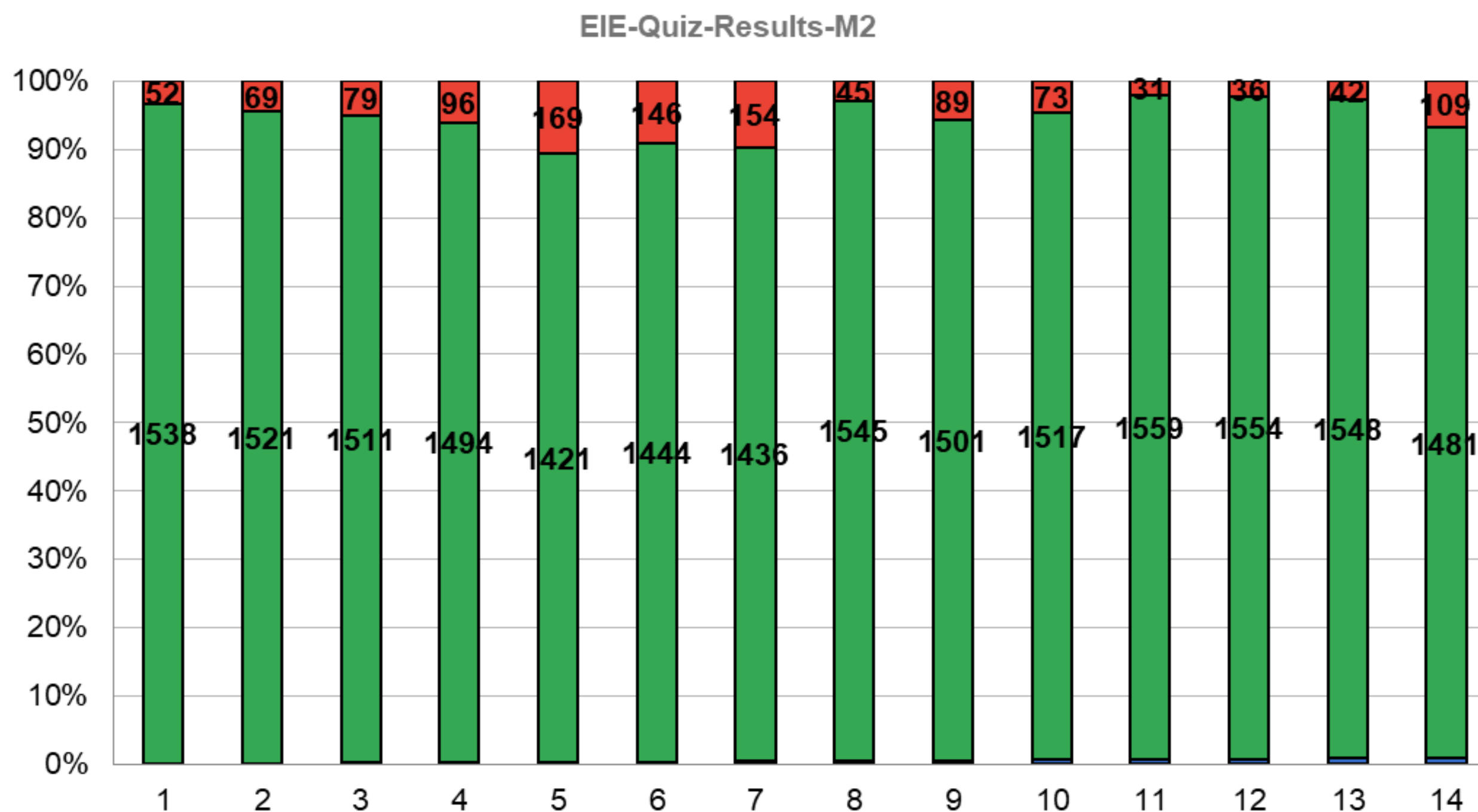
Essentials of Innovation and Entrepreneurship

EIE - I

Module M3 - Opportunity Identification & Ethics

EIE M2 Session – Poll/Mentimeter Results

EIE M2 Session Quiz - Discussion



Accessing EIE 1 Materials

Home

NEFT/RTGS Details

Online Payments

Student Grievance Redressal System

My Courses

Time Table

Bank Consent

My Attendance

Results

Seating Info

Video Archives

My Quizzes

Entrance Exam

Calender

Announcements

My Profile

My Courses > UZ24UZ221A : CIE Essentials of Innovation & Entrepreneurship - Part 1

Course Units

Introduction

Objectives

Outcomes

Outline

Syllabus

References

Introduction to EIE & Entrepreneurial Mindset

Idea to Market, Opportunity, Teams

Value Proposition, Business Model Canvas (BMC), Pitching

Finance, Digital Marketing, AI for Innovation

Class	AV Summary	Live Videos	Slides	Notes	Assignments	QB	QA
M1 - Orientation and Introducti...	1	-	1	-	-	1	-
M2 - Entrepreneurial Mindset	1	-	2	-	-	1	-
M3 - Design Thinking & Effectua...	-	-	-	-	-	-	-

★ Applicable to whole unit - No content

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My Courses

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My Profile

My Courses

Select Semester Sem-3

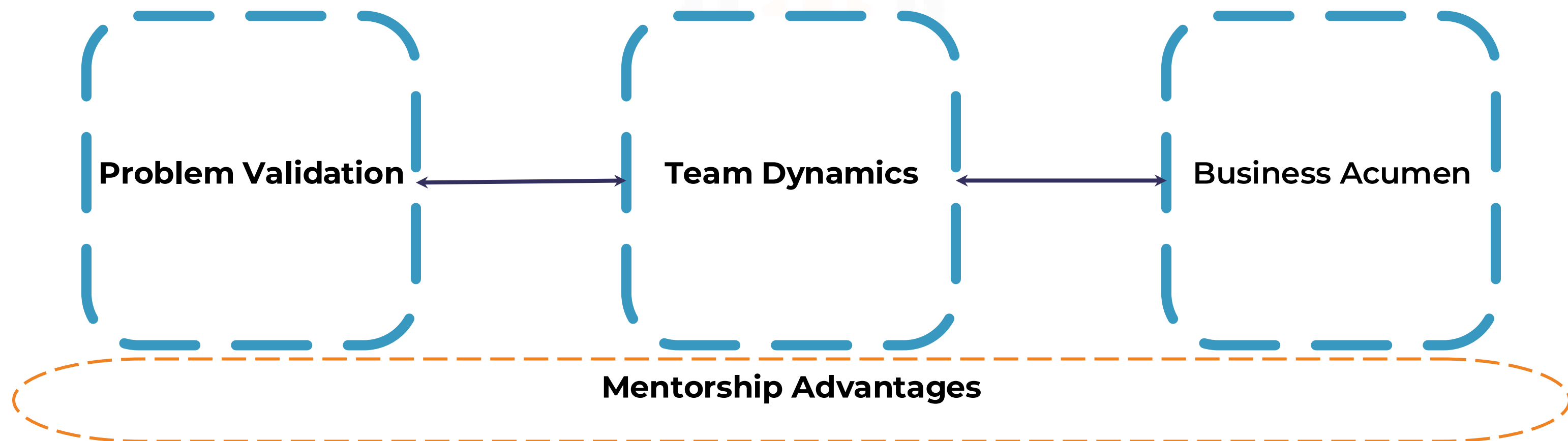
Course Code	Course Title	Course Type	Status	Action
UE24CS242A	Web Technologies	CC	Enrolled	
UE24CS243A	Automata Formal Languages and Logic	CC	Enrolled	
UE24CS251A	Digital Design and Computer Organization	CC	Enrolled	
UE24CS252A	Data Structures and its Applications	CC	Enrolled	
UE24MA242A	Mathematics for Computer Science and Engineering	CC	Enrolled	
UZ24UZ221A	CIE Essentials of Innovation & Entrepreneurship - Part 1	CC	Enrolled	

EIE I - CIE Spark 2025

Validate Your Problem. Forge Your Team. Think Like a Founder.

CIE Spark 2025 is your essential first move: transforming your **raw idea** into a **validated, team-backed venture**, giving you an undeniable advantage for a standout performance in **CIE Ignite**.

The 4 Pillars of CIE Spark 2025



Planned Partnerships

EIE 1 – CIE Spark 2025

"Innovation season is coming. Keep your eyes on CIE Spark!"

EIE M2 Session Quiz - Discussion

5. In the NABC framework, the “A” stands for:

1. Approach – 1421 students got it right!

2. Analysis

3. Assets

4. Adaptability

6. The best startup opportunities generally come from:

a) Problems that customers face and are willing to pay to solve – 1444 students got it right!

b) Copying successful businesses from other countries

c) Trending topics on social media

d) Problems everyone faces but no one will pay for

7. The primary goal of building a Minimum Viable Product (MVP) is to:

a) Launch the cheapest product possible

b) Test core assumptions with real customers– 1436 students got it right!

c) Impress investors with advanced features

d) Beat competitors to market at all costs

3 For Thought

ChatGPT Go (India-only): A new low-cost plan (~₹399/month) that gives you way more usage than free, faster replies, and access to advanced models—plus UPI support. Feels like a broader rollout is on the way.

Perplexity Finance: Rolling out real-time stock tracking (including after-hours) and live earnings-call transcripts for Indian stocks across web and apps. Most features are already available, with Pro/Max and Airtel users typically getting early access and a few extra tools.

Gemma 3 “270M” : It’s a tiny, open-weights model (270M params) that’s super easy to run and tweak, perfect for quick, task-specific fine-tunes and instruction following. Think of it as the ultra-light sibling that rounds out the Gemma 3 lineup below the 1B–27B models.

Today's Topics

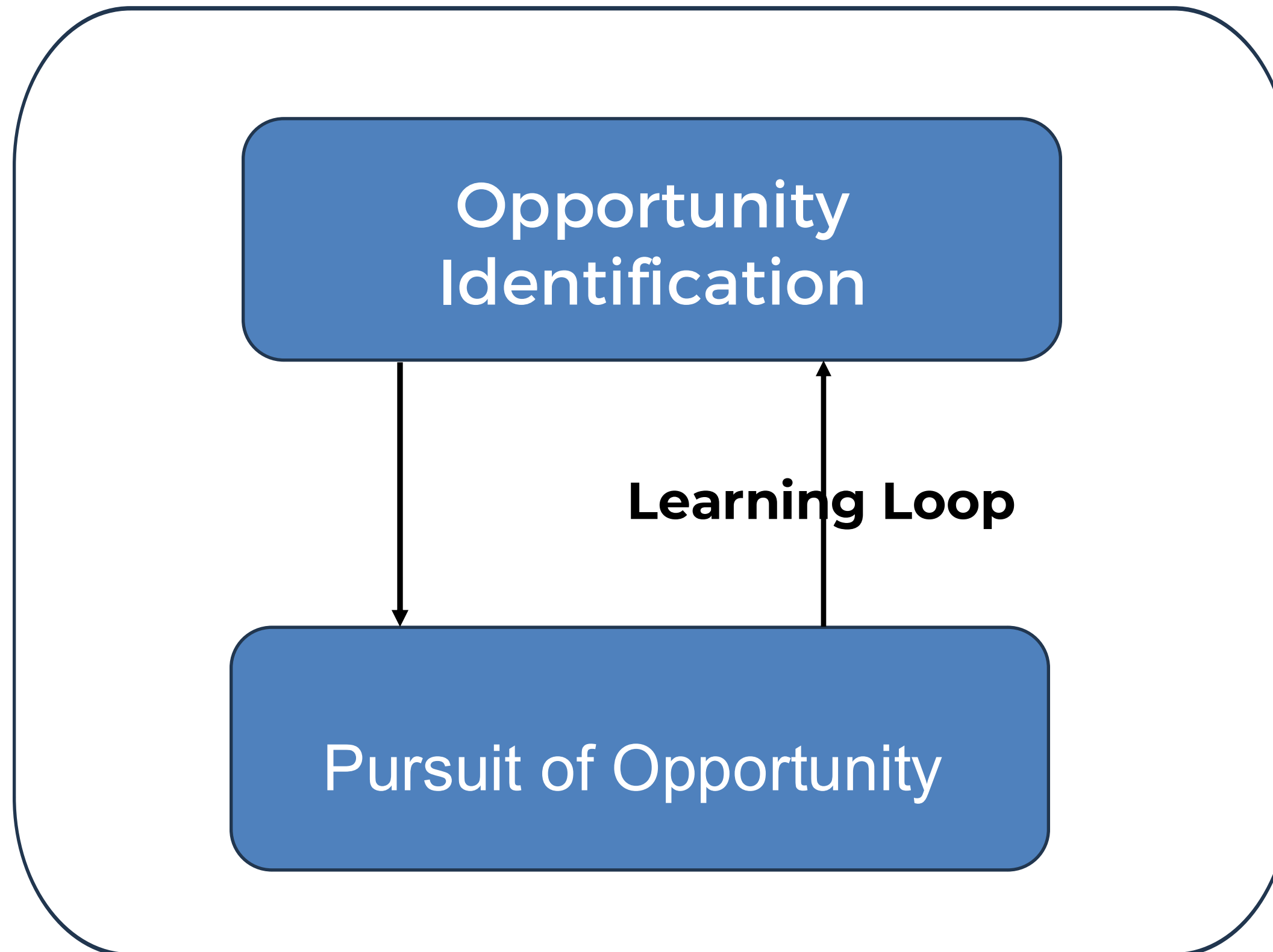


Opportunity Identification

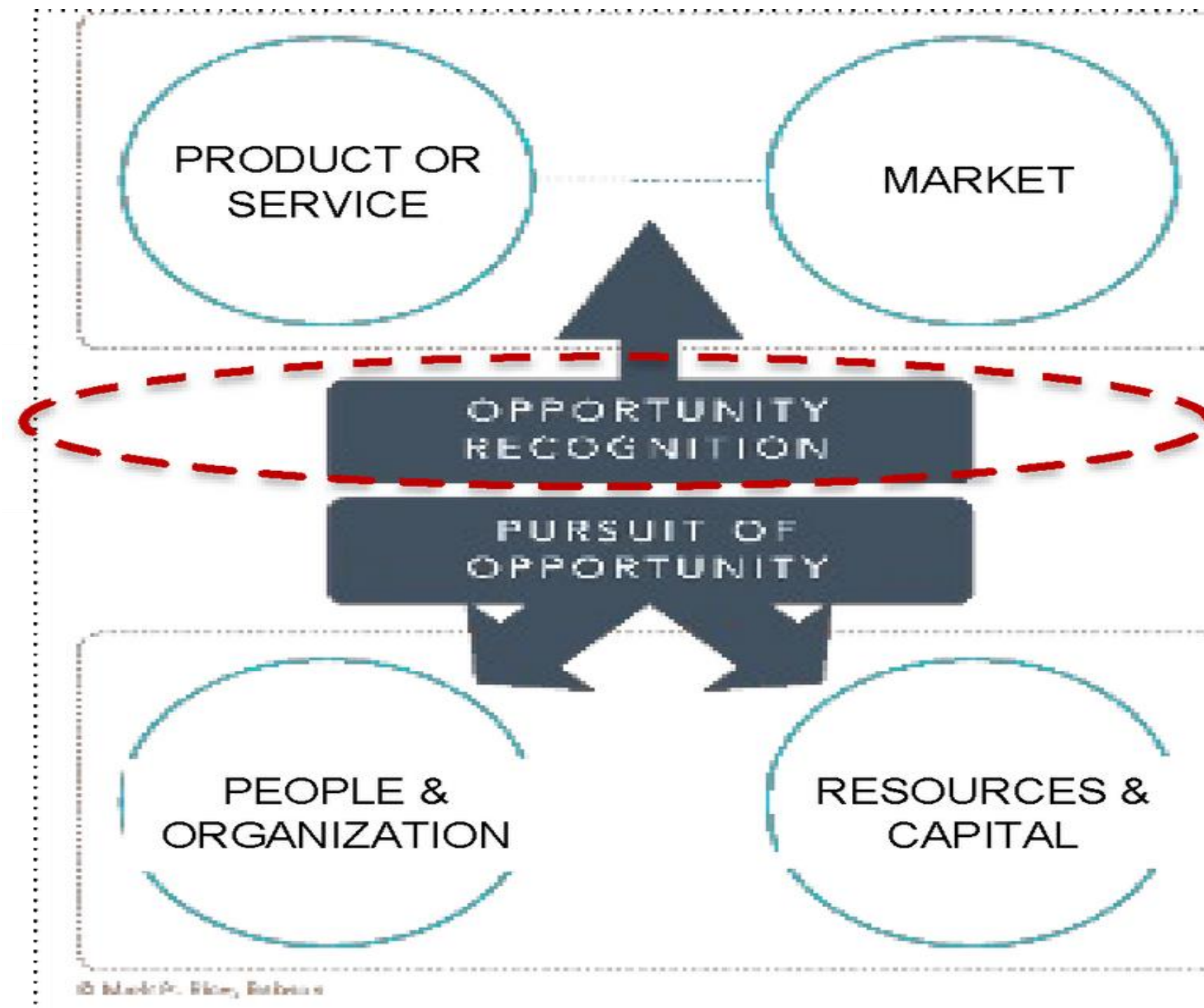


Ethics for Startups

BIG PICTURE: STARTUP (VENTURE)



BIG PICTURE: STARTUP (VENTURE)



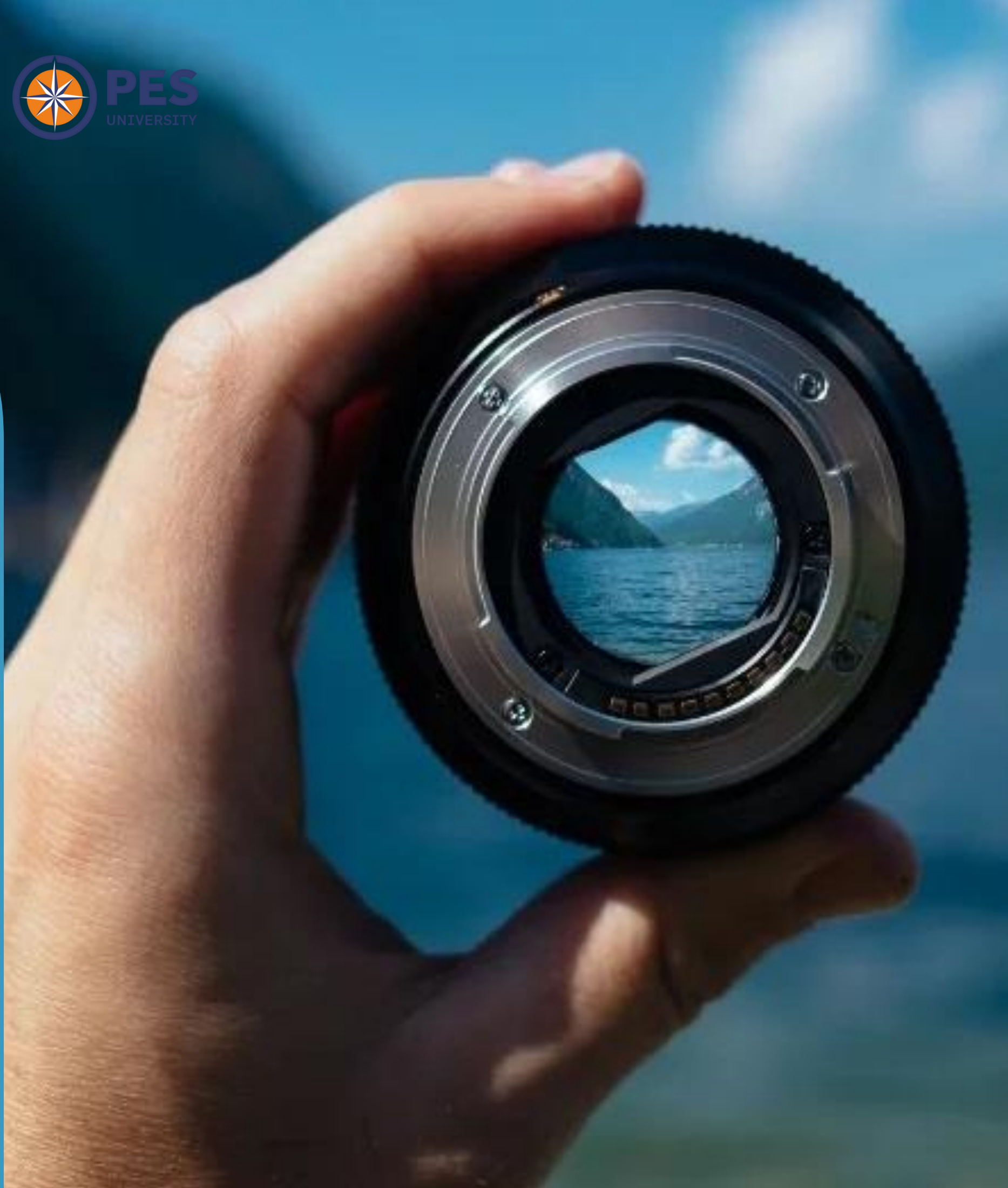
Source: Thomas Byers, Stanford University

Idea to Market Process



Idea to Market Process





Opportunity Identification

Participant Opinion

**Is an *Idea* the same as an
Opportunity?**

Mentimeter

Join at www.menti.com
use the code: 1603 9020



Idea vs. Opportunity

IDEA

The lightbulb moment,
a flash of inspiration that ignites the imagination



Opportunity

The open door, a chance waiting to be seized
or a pathway to creating value



Opportunity Fundamentals

- Great opportunities are often disguised as **difficult problems**
- Potential to create significant **value for the customer**
- **Customer's pain** source for entrepreneur's value creation
- New **technology** may or may not be a biz opportunity
- Shift/changes in regulatory **policy** can be an opportunity
- **Serendipity** (chance) can throw up opportunities 😊

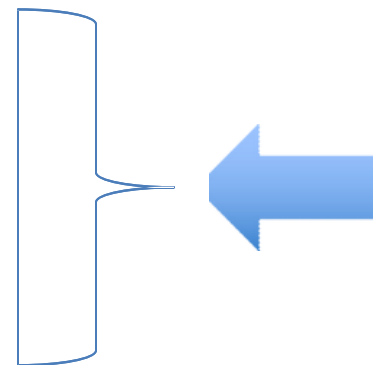
4 Simple Ways to Have a Great Idea (TED Talk, start 45s, total ~5min)



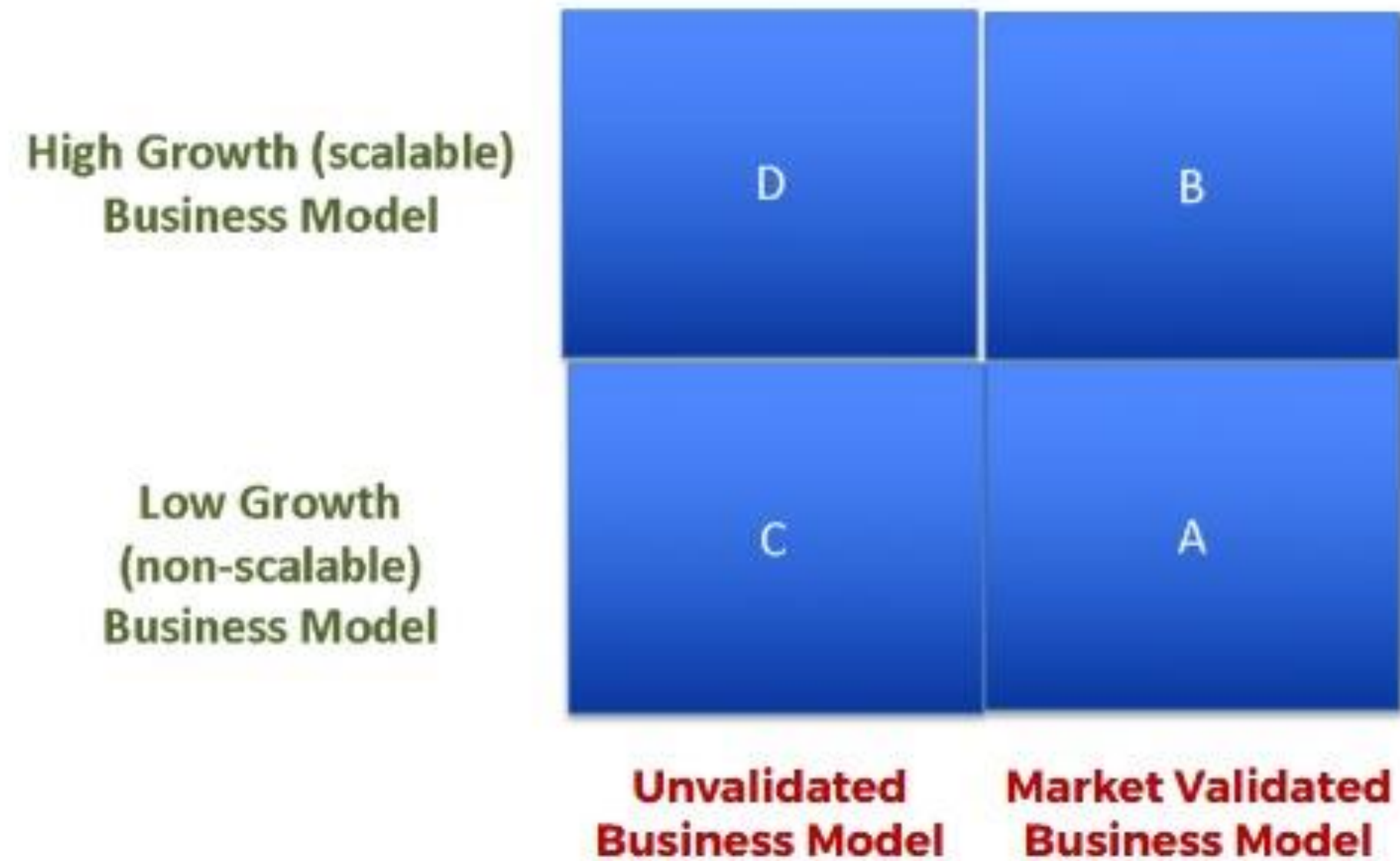
**“Every Problem is an Opportunity”
- Vinod Khosla
(Khosla Ventures & Founder Sun Microsystems)**

Sources of Startup Opportunity

- Customers' **needs** (or changing needs)
- **Personal** (experience of) need or problem
- Changes: **Market**-driven
- Changes: **Technology**-driven
- **Discontinuities** (social, technological, market)
- Role of **Chance/Luck** (“right place, right time”)

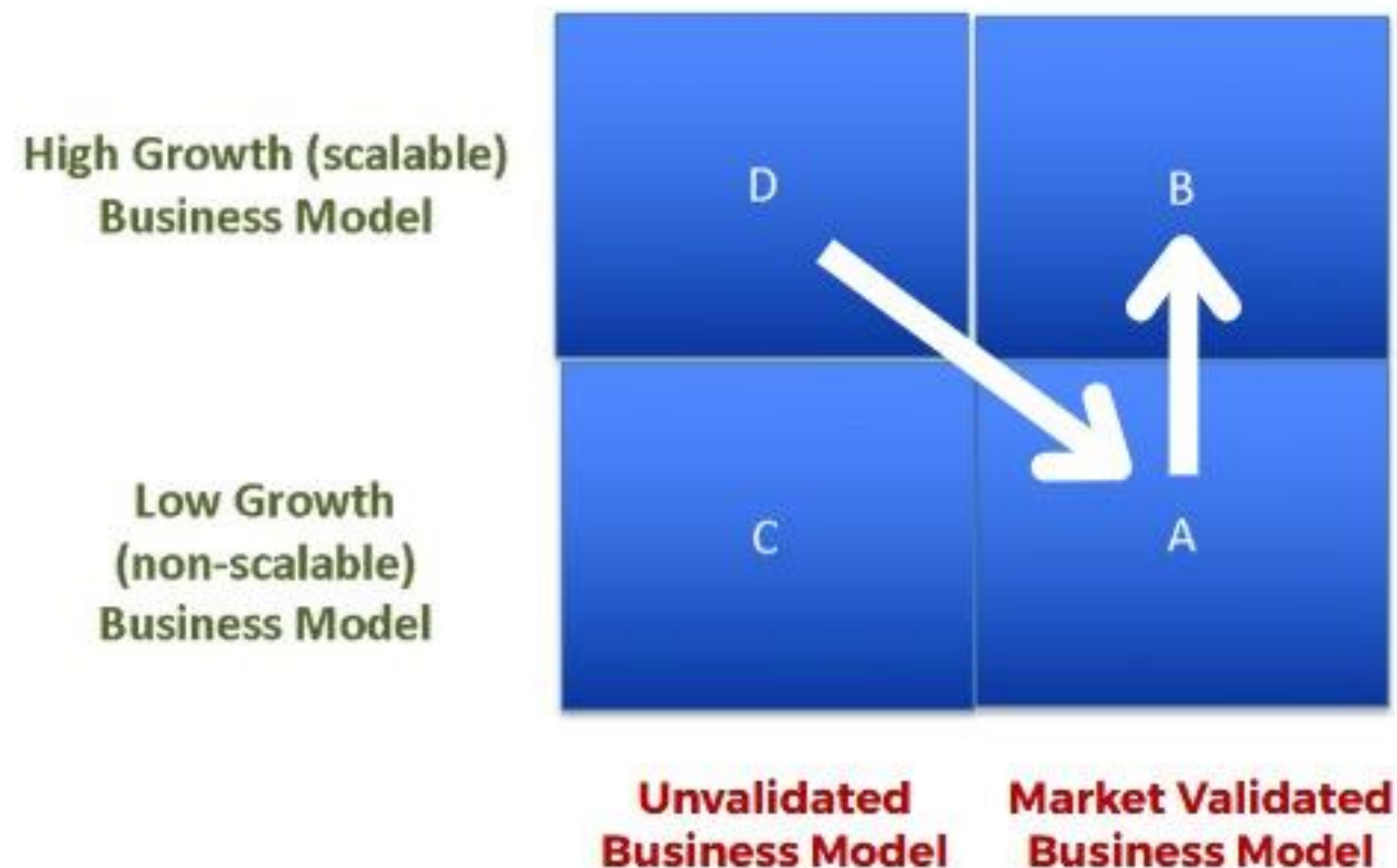


Quiz: IDE* Startup (which quadrant?)



** Innovation-driven entrepreneurship*

Quiz #1: IDE Startup (which quadrant?)



A: Small Business
B: Scaleup
D: Startup
C: No Go!

What makes an Opportunity attractive?



Current
Real Need
or Problem



Problem/Need is
Important
to the customer



Customers'
willing to ***Pay***
& is *likely*
Profitable

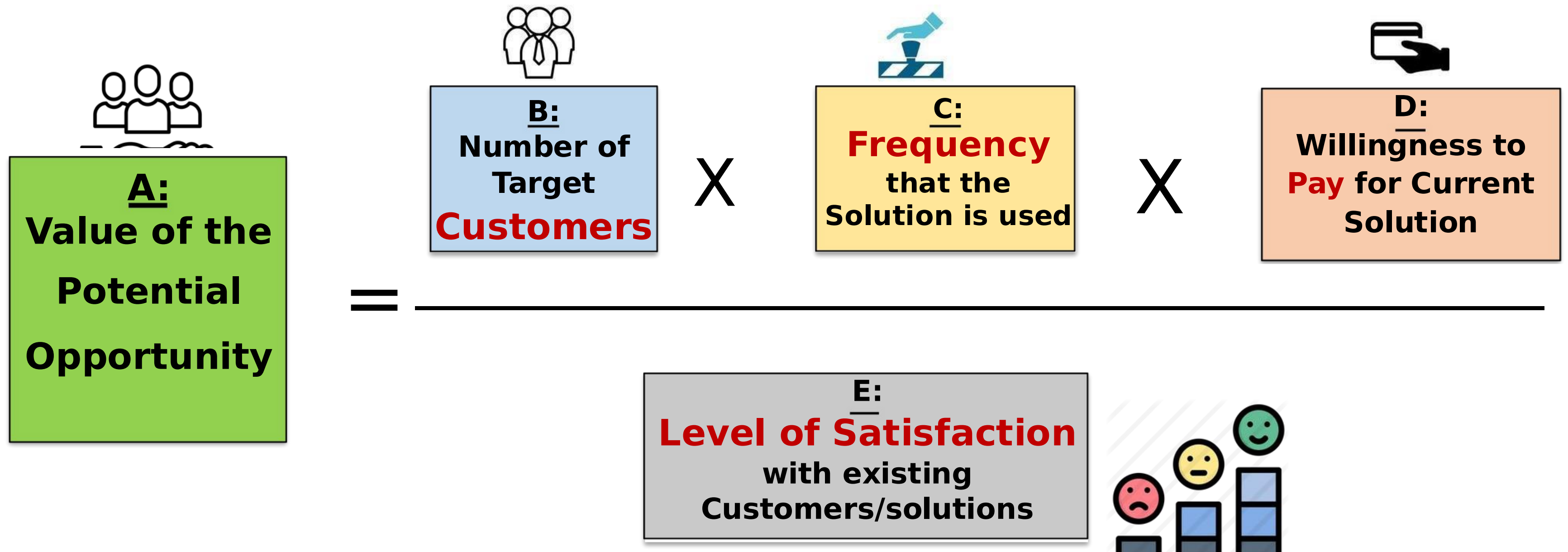


Solvable
in the near future
with resources
(present/acquired)

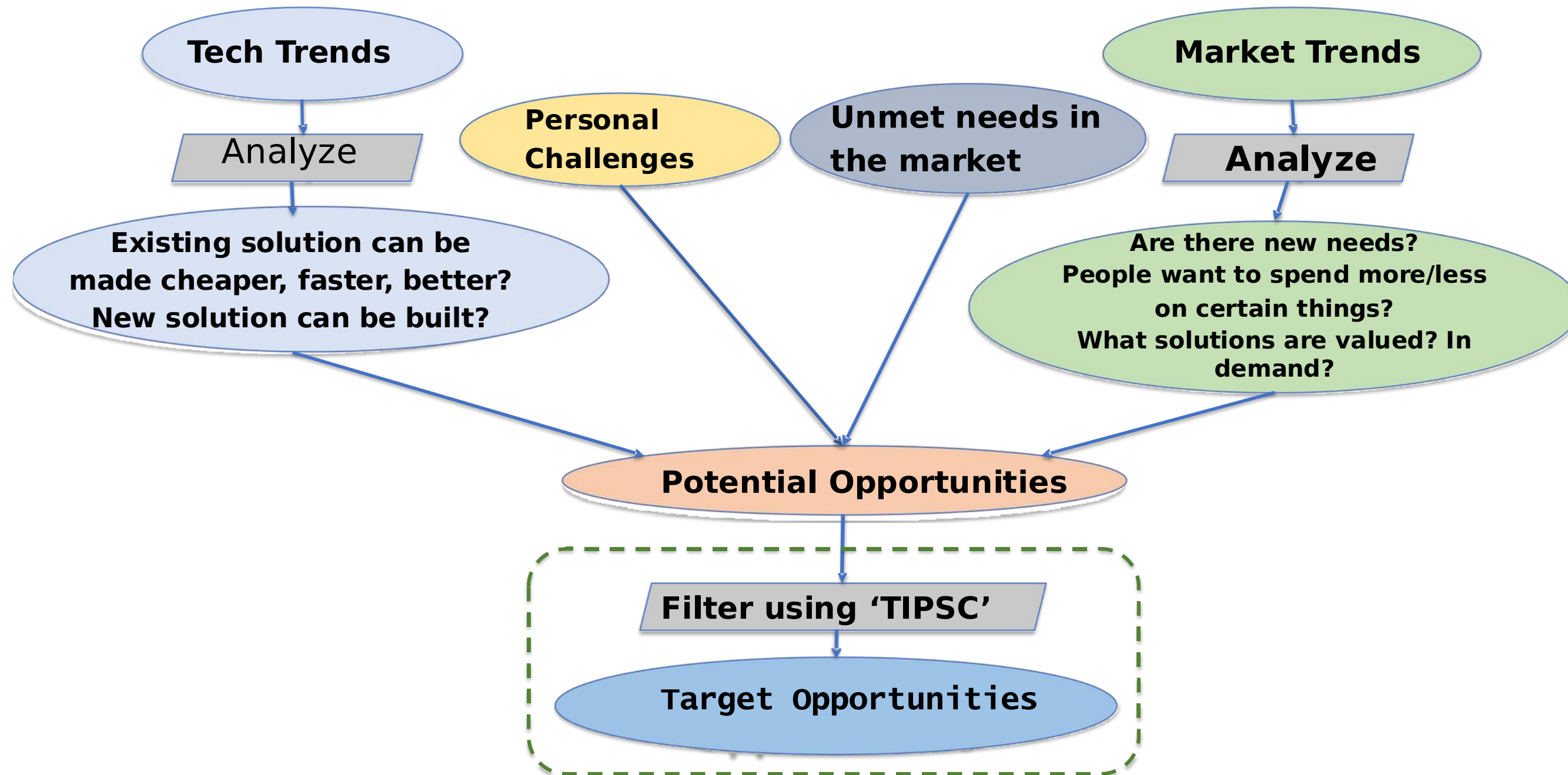


Context
Industry, Market,
Regulations, etc
are understood

'Value' of an Opportunity



Identifying a Good Opportunity Summary (Flowchart)



Summary: Identifying a Good Opportunity

- Problem: Is this a 'good' & real problem (today, not in the distant future)?
- Need: Who needs this (customer) and what benefit is the customer seeking?
- Business: Is this viable from a 'business'?
- Execution: Is there a path for you/team to get started on above?
 - Your team would be an ideal founding team for this opportunity?
 - Testable: Can this problem & solution be tested (from a customer dev perspective)?

Use **TIPSC** to narrow down on opportunity: **Timely, Important, Profitable, Solvable, Context**

Founder Talk

Team Aalap.ai





PERSONALISED MUSIC SO EVERY VIEWER PAYS ATTENTION

WHAT MAKES AALAP UNIQUE?

1. FULLY CUSTOM FOUNDATION MODELS.
2. TRAINED FOR PERFORMANCE-DRIVEN USE (ADS, BRANDED EXPERIENCES)
3. BUILT-IN BRAND + PERSONA ALIGNMENT LOGIC

USE CASES

AD CREATIVES | AUDIO BRANDING | IN-STORE RETAIL MUSIC



Multi Modal Input

Input your music preferences through text, audio, or video, making it easy to create the perfect soundtrack.



Lightning Fast Generation

Generate professional-quality music in seconds, not days or weeks.



Royalty Free Music

Royalty-free music for your projects, ensuring professional quality without licensing fees.



Target Audience Focus

Create music that resonates with your specific demographic and increases engagement.

[AALAP.AI](https://aalap.ai) | TEAM@AALAP.AI

Example: Opportunity identification

Brian Chesky and Joe Gebbia, the founders of Airbnb, noticed a unique opportunity during a design conference in San Francisco. Hotel rooms (which were expensive) were in short supply, and they realized that people had spare rooms in their homes.



This opportunity identification led to the creation of Airbnb, now a global marketplace for accommodations and experiences. Airbnb disrupted the traditional hotel industry by capitalizing on the sharing economy and offering unique and affordable lodging options.

Example: Opportunity identification

S

Patrick and John Collison, founders of Stripe, noticed that the *hardest part of starting an internet business at the time wasn't coming up with an idea, turning the idea into code or getting people to hear about it and pay for it. The hardest part was finding a way to accept customer's money.*

Stripe, now a multinational financial services company with a valuation of about \$65 billion and over \$1 trillion in payment volume processed in 2023. Its customers include the likes of Amazon, IBM, Lyft, NASDAQ and Google.

Example: Opportunity identification



Reed Hastings & Marc Randolph, the co-founders of Netflix, recognized the opportunity to disrupt the video rental industry. They observed the inconvenience of late fees and the limitations of brick-and-mortar rental stores, and identified the potential of a subscription-based model for DVD rentals.

Netflix initially started as a DVD-by-mail rental service (2007), offering subscribers unlimited rentals without late fees. As technology advanced, they shifted their focus to streaming media, recognizing the opportunity to provide on-demand entertainment to customers through the internet.

Post-Session Activity: Problem / Needs

Inspiration: UN SDG Goals



Post-Session Activity: Problem / Needs

Brainstorm/Discuss & list 3-5 biggest (real) **Problems/ Needs** (per category) for 2-3 categories

Sample categories: *Personal Experience, Environment, Entertainment, etc*

An inspiration for categories: [UN Sustainable Development Goals \(SDG\)](#)



Post-Session Activity: Napkin Pitch

- Select the top 2-3 problem/opportunity statements
- Capture this in '***Napkin Pitch***' (on the right) for each problem/opportunity statement
- Use the ***Napkin pitch*** (per opportunity) as a starting point to validate the opportunity

<u>The Big Idea</u> (Opportunity Statement)	<u>Needs/ Benefits</u> (what your solution can deliver to Customers/Users)
<u>Execution</u> (what does it take to implement this solution?)	<u>Viable Business?</u> (worth the effort? What are the top risks and key assumptions behind this?)



MORALITY



RESPONSIBILITY



TRUST



RELIABILITY

BUSINESS ETHICS



BEHAVIOR



CHOICE



RELATIONSHIP



PRINCIPLE

Ethics (For Startups)

What is Ethics?

What is Business Ethics?

[Video](#) (~3mins)

Ethics and Business

- Ethics is concerned with doing the **right (moral)** thing
- It is a solid foundation on which **integrity** and **reputation** can be built
- Integrity and reputation is a must for **long-term business** success
- While the above is true any enterprise, it is especially important for startups as this can bring down the entire effort if one is not mindful of ethics (**Why?**)

Downward Spiral of Unethical Practices in Business

Pressure to win: intense competition, funding stress, urge to scale fast.

First temptation: bending small rules (exaggeration, overselling).

Shifting moral compass: rationalizing bigger compromises.

Spiral deepens: shortcuts, cover-ups, culture of ignoring ethics.

Consequences: loss of trust, legal trouble, collapse (Enron, Theranos).

Founder's role: set non-negotiable values; ethics = survival strategy.



Redefining Entrepreneurship

“Entrepreneurship is when you
act [**ethically**]

upon opportunities and ideas and transform them into value for
others”

Ethical Leadership Mindset

Integrity

Ethical leader consistently demonstrate honesty, trustworthiness, and a strong moral compass.

Fairness & Justice

Ethical leaders treat all individuals with fairness and impartially.

Ethical Leadership Mindset

Decision Making

Ethical leaders prioritize ethical considerations in their decision making process.

Responsibility & Accountability

Ethical leaders take responsibility for their actions and the outcomes of their decisions.

Overlooking Ethics Results in ... Scandals

Corporate Examples, Global/Indian

- Enron, Volkswagen, Lehman Brothers, Facebook [list](#)
- Satyam Computers, IL&FS [list](#)

Startup Examples, Global/Indian

- [Theranos - A Fallen Unicorn](#)
- [GoMechanic](#)

GoMechanic fraud: How chasing growth at all costs brought down a rising startup

GoMechanic had all the right mix to be a pioneer in the garage services business. It had the first-mover advantage, IIM A-graduate founders, big names of the VC world either invested or interested, a strong customer base, and a network of nearly 1,000 garages across 40 cities in place. Yet things went downhill.

Ethics-related Issues can arise in Small/Large & Local/Global enterprises

Obstacles to Ethical Decision-Making

- **Complacency:** We believe that “it can’t happen here.”
- **Self-Delusion:** We judge ourselves by our intentions, while others judge us by our actions
- **Rationalizations:** We make up justifications and excuses for ethical missteps
- **Survival Mentality:** We convince ourselves that ethical missteps are necessary



How to Build a Ethical Startup

[Video](#)

(Stanford University ~2mins)

Session Feedback

Mentimeter

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Data Ethics

"Data Ethics encompasses the moral obligations of gathering, protecting and using personally identifiable information and how it affects individuals"

5 Key Aspects of Data Ethics:

- **Ownership**
- **Transparency**
- **Privacy**
- **Intention**
- **Outcomes**



Session Summary - Opportunity

Trends – Spot emerging technologies, cultural shifts, and lifestyle changes shaping demand.

Industry Gaps – Look for inefficiencies, unmet needs, or outdated practices in specific sectors.

Problems – Identify everyday frustrations, pain points, or bottlenecks worth solving.

Shifts – Track regulatory, demographic, or behavioral changes creating new opportunities.

Customer Needs – Listen actively to users; validate what they want (and what current solutions miss).

👉 Together, Frameworks like **TIPSC** helps filter raw ideas into real, high-potential opportunities.

Session Summary - Ethics

Integrity & Transparency – Build trust by being honest with customers, employees, and partners.

Fair Practices – Avoid exploitation, ensure fair wages, safe work conditions, and inclusive opportunities.

Sustainability – Balance profit with responsibility toward environment and society.

Accountability – Take ownership for actions; avoid cutting corners for short-term gains.

Privacy & Consent – Collect and use personal data responsibly, with clear user permission.

Bias & Fairness – Prevent algorithmic discrimination; ensure equitable outcomes across groups.

Security – Safeguard data against breaches, misuse, and unauthorized access.

Transparency in AI – Clearly explain how data is used and decisions are made.

👉 Together, **business ethics + data ethics = long-term trust, brand value, and resilience.**

Why it's a **Bad Idea** to Invest in 'Good' Ideas!



Courtesy: Alberto Savoia (ex-Google)

Caselet: Nespresso (HBR Video)



Sketch Out Your Hypotheses

Categories of Opportunity

- 1. Increasing the value of a product or service**
- 2. New applications of existing means or technologies**
- 3. Creating mass markets**
- 4. Customization for individuals**
- 5. Increasing reach (allowing access to more customers)**
- Managing the supply chain**
- 7. Convergence of industries**
- 8. Process innovation**
- 9. Increasing the scale of the firm**

Session Quiz

Please take the Session Quiz on PESU Academy now

Thank You!